Modeling Neutron Stars
Lecture-9

Chiranjib Mondal
LPC, Université Caen Normandie
https://sites.google.com/view/nucleartheory/home

UNIVERSITÉ CAEN NORMANDIE

Nuclear Theory Group
February 27, 2021

Contents:

Elementary General Theory of Relativity

- Recap: Minkowski Metric
- General Metric
- Principle of equivalence

Flat space time with STR

c=1, $g_{\mu\nu} \rightarrow \eta_{\mu\nu}$

- Lorentz transformation leaves the *proper time* invariant

$$d\tau^2 = dt^2 - dx^2 - dy^2 - dz^2 = 0$$

Flat space time with STR

c=1, $g_{\mu\nu} \rightarrow \eta_{\mu\nu}$

- Lorentz transformation leaves the *proper time* invariant

$$
\begin{aligned}
d\tau^2 &= dt^2 - dx^2 - dy^2 - dz^2 = 0 \\
\Rightarrow d\tau &= \sqrt{1 - \left(\frac{d\boldsymbol{x}}{dt}\right)^2}\, dt = 0
\end{aligned}
$$

Flat space time with STR

c=1, $g_{\mu\nu} \rightarrow \eta_{\mu\nu}$

- Lorentz transformation leaves the *proper time* invariant

$$
\begin{aligned}
d\tau^2 &= dt^2 - dx^2 - dy^2 - dz^2 = 0 \\
\Rightarrow d\tau &= \sqrt{1 - \left(\frac{d\boldsymbol{x}}{dt}\right)^2}\, dt = 0 \Rightarrow \left|\frac{d\boldsymbol{x}}{dt}\right| = c = 1
\end{aligned}
$$

Flat space time with STR

c=1, $g_{\mu\nu} \to \eta_{\mu\nu}$

- Lorentz transformation leaves the *proper time* invariant

$$
\begin{aligned}
d\tau^2 &= dt^2 - dx^2 - dy^2 - dz^2 = 0 \\
\Rightarrow d\tau &= \sqrt{1 - \left(\frac{d\boldsymbol{x}}{dt}\right)^2}\, dt = 0 \Rightarrow \left|\frac{d\boldsymbol{x}}{dt}\right| = c = 1 \\
\text{With metric, } \eta_{\mu\nu} &= \mathrm{diag}(1, -1, -1, -1)
\end{aligned}
$$

Flat space time with STR

c=1, $g_{\mu\nu} \to \eta_{\mu\nu}$

- Lorentz transformation leaves the *proper time* invariant

$$
\begin{aligned}
d\tau^2 &= dt^2 - dx^2 - dy^2 - dz^2 = 0 \\
\Rightarrow d\tau &= \sqrt{1 - \left(\frac{d\boldsymbol{x}}{dt}\right)^2}\, dt = 0 \Rightarrow \left|\frac{d\boldsymbol{x}}{dt}\right| = c = 1 \\
\text{With metric, } \eta_{\mu\nu} &= \mathrm{diag}(1, -1, -1, -1) \\
\Rightarrow d\tau^2 &= \eta_{\mu\nu}\, dx^\mu\, dx^\nu
\end{aligned}
$$

Flat space time with STR

c=1, $g_{\mu\nu} \to \eta_{\mu\nu}$

- Lorentz transformation leaves the *proper time* invariant

$$
\begin{aligned}
d\tau^2 &= dt^2 - dx^2 - dy^2 - dz^2 = 0 \\
\Rightarrow d\tau &= \sqrt{1 - \left(\frac{d\boldsymbol{x}}{dt}\right)^2}\, dt = 0 \Rightarrow \left|\frac{d\boldsymbol{x}}{dt}\right| = c = 1 \\
\text{With metric, } \eta_{\mu\nu} &= \mathrm{diag}(1, -1, -1, -1) \\
\Rightarrow d\tau^2 &= \eta_{\mu\nu}\, dx^\mu\, dx^\nu = \eta_{\mu\nu}\, dx'^\mu\, dx'^\nu
\end{aligned}
$$

Flat space time with STR

c=1, $g_{\mu\nu} \to \eta_{\mu\nu}$

- Lorentz transformation leaves the *proper time* invariant

$$\begin{aligned} d\tau^2 &= dt^2 - dx^2 - dy^2 - dz^2 = 0 \\ \Rightarrow d\tau &= \sqrt{1 - \left(\frac{d\boldsymbol{x}}{dt}\right)^2}\, dt = 0 \Rightarrow \left|\frac{d\boldsymbol{x}}{dt}\right| = c = 1 \\ \text{With metric, } \eta_{\mu\nu} &= \text{diag}(1, -1, -1, -1) \\ \Rightarrow d\tau^2 &= \eta_{\mu\nu}\, dx^\mu\, dx^\nu = \eta_{\mu\nu}\, dx'^\mu\, dx'^\nu \end{aligned}$$

- Co-ordinates transform under Lorentz transformation:

$$x^\mu \to x'^\mu = \Lambda^\mu{}_\nu x^\nu$$

Flat space time with STR

c=1, $g_{\mu\nu} \to \eta_{\mu\nu}$

- Lorentz transformation leaves the *proper time* invariant

$$\begin{aligned} d\tau^2 &= dt^2 - dx^2 - dy^2 - dz^2 = 0 \\ \Rightarrow d\tau &= \sqrt{1 - \left(\frac{d\boldsymbol{x}}{dt}\right)^2}\, dt = 0 \Rightarrow \left|\frac{d\boldsymbol{x}}{dt}\right| = c = 1 \\ \text{With metric, } \eta_{\mu\nu} &= \text{diag}(1, -1, -1, -1) \\ \Rightarrow d\tau^2 &= \eta_{\mu\nu}\, dx^\mu\, dx^\nu = \eta_{\mu\nu}\, dx'^\mu\, dx'^\nu \end{aligned}$$

- Co-ordinates transform under Lorentz transformation:

$$\begin{aligned} x^\mu \to x'^\mu &= \Lambda^\mu{}_\nu x^\nu \\ \Rightarrow \eta_{\alpha\beta}\, dx^\alpha\, dx^\beta &= \eta_{\mu\nu}\, dx'^\mu\, dx'^\nu \end{aligned}$$

Flat space time with STR

c=1, $g_{\mu\nu} \to \eta_{\mu\nu}$

- Lorentz transformation leaves the *proper time* invariant

$$\begin{aligned} d\tau^2 &= dt^2 - dx^2 - dy^2 - dz^2 = 0 \\ \Rightarrow d\tau &= \sqrt{1 - \left(\frac{d\boldsymbol{x}}{dt}\right)^2}\, dt = 0 \Rightarrow \left|\frac{d\boldsymbol{x}}{dt}\right| = c = 1 \\ \text{With metric, } \eta_{\mu\nu} &= \text{diag}(1, -1, -1, -1) \\ \Rightarrow d\tau^2 &= \eta_{\mu\nu}\, dx^\mu\, dx^\nu = \eta_{\mu\nu}\, dx'^\mu\, dx'^\nu \end{aligned}$$

- Co-ordinates transform under Lorentz transformation:

$$\begin{aligned} x^\mu \to x'^\mu &= \Lambda^\mu{}_\nu x^\nu \\ \Rightarrow \eta_{\alpha\beta}\, dx^\alpha\, dx^\beta &= \eta_{\mu\nu}\, dx'^\mu\, dx'^\nu = \eta_{\mu\nu} \Lambda^\mu{}_\alpha \Lambda^\nu{}_\beta\, dx^\alpha\, dx^\beta \end{aligned}$$

Flat space time with STR

c=1, $g_{\mu\nu} \to \eta_{\mu\nu}$

- Lorentz transformation leaves the *proper time* invariant

$$\begin{aligned} d\tau^2 &= dt^2 - dx^2 - dy^2 - dz^2 = 0 \\ \Rightarrow d\tau &= \sqrt{1 - \left(\frac{d\boldsymbol{x}}{dt}\right)^2}\, dt = 0 \Rightarrow \left|\frac{d\boldsymbol{x}}{dt}\right| = c = 1 \\ \text{With metric, } \eta_{\mu\nu} &= \text{diag}(1, -1, -1, -1) \\ \Rightarrow d\tau^2 &= \eta_{\mu\nu}\, dx^\mu\, dx^\nu = \eta_{\mu\nu}\, dx'^\mu\, dx'^\nu \end{aligned}$$

- Co-ordinates transform under Lorentz transformation:

$$\begin{aligned} x^\mu \to x'^\mu &= \Lambda^\mu{}_\nu x^\nu \\ \Rightarrow \eta_{\alpha\beta}\, dx^\alpha\, dx^\beta &= \eta_{\mu\nu}\, dx'^\mu\, dx'^\nu = \eta_{\mu\nu} \Lambda^\mu{}_\alpha \Lambda^\nu{}_\beta\, dx^\alpha\, dx^\beta \\ \Rightarrow \eta_{\alpha\beta} &= \eta_{\mu\nu} \Lambda^\mu{}_\alpha \Lambda^\nu{}_\beta \end{aligned}$$

Boost

$\eta_{\alpha\beta} = \eta_{\mu\nu}\Lambda^{\mu}{}_{\alpha}\Lambda^{\nu}{}_{\beta}$

- A boost along x-axis,

$$
\begin{aligned}
x'^0 &= \Lambda^0{}_0 x^0 + \Lambda^0{}_1 x^1 \\
x'^1 &= \Lambda^1{}_0 x^0 + \Lambda^1{}_1 x^1 \\
x'^2 &= x^2 \\
x'^3 &= x^3
\end{aligned}
$$

Boost

$\eta_{\alpha\beta} = \eta_{\mu\nu}\Lambda^{\mu}{}_{\alpha}\Lambda^{\nu}{}_{\beta}$

- A boost along x-axis,

$$\begin{aligned} x'^0 &= \Lambda^0{}_0 x^0 + \Lambda^0{}_1 x^1 \\ x'^1 &= \Lambda^1{}_0 x^0 + \Lambda^1{}_1 x^1 \\ x'^2 &= x^2 \\ x'^3 &= x^3 \end{aligned}$$

- From the metric relation,

$$\begin{aligned} 1 &= \left(\Lambda^0{}_0\right)^2 - \left(\Lambda^1{}_0\right)^2 = -\left[\left(\Lambda^0{}_1\right)^2 - \left(\Lambda^1{}_1\right)^2\right] \\ 0 &= \Lambda^0{}_0\Lambda^0{}_1 - \Lambda^1{}_0\Lambda^1{}_1 \end{aligned}$$

Boost

$\eta_{\alpha\beta} = \eta_{\mu\nu}\Lambda^{\mu}{}_{\alpha}\Lambda^{\nu}{}_{\beta}$

- A boost along x-axis,

$$\begin{aligned} x'^0 &= \Lambda^0{}_0 x^0 + \Lambda^0{}_1 x^1 \\ x'^1 &= \Lambda^1{}_0 x^0 + \Lambda^1{}_1 x^1 \\ x'^2 &= x^2 \\ x'^3 &= x^3 \end{aligned}$$

- From the metric relation,

$$\begin{aligned} 1 &= \left(\Lambda^0{}_0\right)^2 - \left(\Lambda^1{}_0\right)^2 = -\left[\left(\Lambda^0{}_1\right)^2 - \left(\Lambda^1{}_1\right)^2\right] \\ 0 &= \Lambda^0{}_0\Lambda^0{}_1 - \Lambda^1{}_0\Lambda^1{}_1 \\ x'^1 &= 0 = \Lambda^1{}_0 x^0 + \Lambda^1{}_1 x^1 \end{aligned}$$

Boost

$\eta_{\alpha\beta} = \eta_{\mu\nu}\Lambda^{\mu}{}_{\alpha}\Lambda^{\nu}{}_{\beta}$

- A boost along x-axis,

$$\begin{aligned} x'^0 &= \Lambda^0{}_0 x^0 + \Lambda^0{}_1 x^1 \\ x'^1 &= \Lambda^1{}_0 x^0 + \Lambda^1{}_1 x^1 \\ x'^2 &= x^2 \\ x'^3 &= x^3 \end{aligned}$$

- From the metric relation,

$$\begin{aligned} 1 &= \left(\Lambda^0{}_0\right)^2 - \left(\Lambda^1{}_0\right)^2 = -\left[\left(\Lambda^0{}_1\right)^2 - \left(\Lambda^1{}_1\right)^2\right] \\ 0 &= \Lambda^0{}_0\Lambda^0{}_1 - \Lambda^1{}_0\Lambda^1{}_1 \\ x'^1 &= 0 = \Lambda^1{}_0 x^0 + \Lambda^1{}_1 x^1 = \left(\Lambda^1{}_0 + \Lambda^1{}_1 v\right) t \end{aligned}$$

Boost

$\eta_{\alpha\beta} = \eta_{\mu\nu}\Lambda^{\mu}{}_{\alpha}\Lambda^{\nu}{}_{\beta}$

- A boost along x-axis,

$$\begin{aligned} x'^0 &= \Lambda^0{}_0 x^0 + \Lambda^0{}_1 x^1 \\ x'^1 &= \Lambda^1{}_0 x^0 + \Lambda^1{}_1 x^1 \\ x'^2 &= x^2 \\ x'^3 &= x^3 \end{aligned}$$

- From the metric relation,

$$\begin{aligned} 1 &= \left(\Lambda^0{}_0\right)^2 - \left(\Lambda^1{}_0\right)^2 = -\left[\left(\Lambda^0{}_1\right)^2 - \left(\Lambda^1{}_1\right)^2\right] \\ 0 &= \Lambda^0{}_0\Lambda^0{}_1 - \Lambda^1{}_0\Lambda^1{}_1 \\ x'^1 &= 0 = \Lambda^1{}_0 x^0 + \Lambda^1{}_1 x^1 = \left(\Lambda^1{}_0 + \Lambda^1{}_1 v\right) t \\ \Rightarrow \Lambda^1{}_0 &= -\Lambda^1{}_1 v \end{aligned}$$

Boost

$\eta_{\alpha\beta} = \eta_{\mu\nu}\Lambda^{\mu}{}_{\alpha}\Lambda^{\nu}{}_{\beta}$

- A boost along x-axis,

$$\begin{aligned} x'^0 &= \Lambda^0{}_0 x^0 + \Lambda^0{}_1 x^1 \\ x'^1 &= \Lambda^1{}_0 x^0 + \Lambda^1{}_1 x^1 \\ x'^2 &= x^2 \\ x'^3 &= x^3 \end{aligned}$$

- So the transformations are,

$$\begin{aligned} \Lambda^0{}_0 &= \Lambda^1{}_1 = \gamma;\ \gamma = (1-v^2)^{-1/2} \\ \Lambda^1{}_0 &= \Lambda^0{}_1 = -v\gamma \\ \Lambda^2{}_2 &= \Lambda^3{}_3 = 1 \end{aligned}$$

- From the metric relation,

$$\begin{aligned} 1 &= \left(\Lambda^0{}_0\right)^2 - \left(\Lambda^1{}_0\right)^2 = -\left[\left(\Lambda^0{}_1\right)^2 - \left(\Lambda^1{}_1\right)^2\right] \\ 0 &= \Lambda^0{}_0\Lambda^0{}_1 - \Lambda^1{}_0\Lambda^1{}_1 \\ x'^1 &= 0 = \Lambda^1{}_0 x^0 + \Lambda^1{}_1 x^1 = \left(\Lambda^1{}_0 + \Lambda^1{}_1 v\right) t \\ \Rightarrow \Lambda^1{}_0 &= -\Lambda^1{}_1 v \end{aligned}$$

Boost

$\eta_{\alpha\beta} = \eta_{\mu\nu}\Lambda^{\mu}{}_{\alpha}\Lambda^{\nu}{}_{\beta}$

- A boost along x-axis,

$$\begin{aligned} x'^0 &= \Lambda^0{}_0 x^0 + \Lambda^0{}_1 x^1 \\ x'^1 &= \Lambda^1{}_0 x^0 + \Lambda^1{}_1 x^1 \\ x'^2 &= x^2 \\ x'^3 &= x^3 \end{aligned}$$

- So the transformations are,

$$\begin{aligned} \Lambda^0{}_0 &= \Lambda^1{}_1 = \gamma;\ \gamma = (1-v^2)^{-1/2} \\ \Lambda^1{}_0 &= \Lambda^0{}_1 = -v\gamma \\ \Lambda^2{}_2 &= \Lambda^3{}_3 = 1 \end{aligned}$$

- From the metric relation,

$$\begin{aligned} 1 &= \left(\Lambda^0{}_0\right)^2 - \left(\Lambda^1{}_0\right)^2 = -\left[\left(\Lambda^0{}_1\right)^2 - \left(\Lambda^1{}_1\right)^2\right] \\ 0 &= \Lambda^0{}_0\Lambda^0{}_1 - \Lambda^1{}_0\Lambda^1{}_1 \\ x'^1 &= 0 = \Lambda^1{}_0 x^0 + \Lambda^1{}_1 x^1 = \left(\Lambda^1{}_0 + \Lambda^1{}_1 v\right) t \\ \Rightarrow \Lambda^1{}_0 &= -\Lambda^1{}_1 v \end{aligned}$$

- This can be generalized for $\boldsymbol{v}$,

$$\begin{aligned} \Lambda^0{}_0 &= \gamma \\ \Lambda^0{}_j &= \Lambda^j{}_0 = -v^j\gamma \\ \Lambda^j{}_k &= \Lambda^k{}_j = \delta^j_k + (\gamma-1)v^j v^k \Big/ \boldsymbol{v}^2 \end{aligned}$$

Boost

$\eta_{\alpha\beta} = \eta_{\mu\nu}\Lambda^{\mu}{}_{\alpha}\Lambda^{\nu}{}_{\beta}$

- A boost along x-axis,

$$\begin{aligned} x'^0 &= \Lambda^0{}_0 x^0 + \Lambda^0{}_1 x^1 \\ x'^1 &= \Lambda^1{}_0 x^0 + \Lambda^1{}_1 x^1 \\ x'^2 &= x^2 \\ x'^3 &= x^3 \end{aligned}$$

- So the transformations are,

$$\begin{aligned} \Lambda^0{}_0 &= \Lambda^1{}_1 = \gamma;\ \gamma = (1-v^2)^{-1/2} \\ \Lambda^1{}_0 &= \Lambda^0{}_1 = -v\gamma \\ \Lambda^2{}_2 &= \Lambda^3{}_3 = 1 \end{aligned}$$

- From the metric relation,

$$\begin{aligned} 1 &= \left(\Lambda^0{}_0\right)^2 - \left(\Lambda^1{}_0\right)^2 = -\left[\left(\Lambda^0{}_1\right)^2 - \left(\Lambda^1{}_1\right)^2\right] \\ 0 &= \Lambda^0{}_0\Lambda^0{}_1 - \Lambda^1{}_0\Lambda^1{}_1 \\ x'^1 &= 0 = \Lambda^1{}_0 x^0 + \Lambda^1{}_1 x^1 = \left(\Lambda^1{}_0 + \Lambda^1{}_1 v\right) t \\ \Rightarrow \Lambda^1{}_0 &= -\Lambda^1{}_1 v \end{aligned}$$

- This can be generalized for $\boldsymbol{v}$,

$$\begin{aligned} \Lambda^0{}_0 &= \gamma \\ \Lambda^0{}_j &= \Lambda^j{}_0 = -v^j\gamma \\ \Lambda^j{}_k &= \Lambda^k{}_j = \delta^j_k + (\gamma-1)v^j v^k \Big/ \boldsymbol{v}^2 \end{aligned}$$

- So for the inverse transformation $x^\mu = \Lambda_\nu{}^\mu x'^\nu$,

$$\begin{aligned} \Lambda_0{}^0 &= \gamma \\ \Lambda_0{}^j &= \Lambda_j{}^0 = v^j\gamma \\ \Lambda_j{}^k &= \Lambda_k{}^j = \delta^j_k + (\gamma-1)v^j v^k \Big/ \boldsymbol{v}^2 \end{aligned}$$

Covariant vectors

Perfect fluid

- *Four-velocity* is defined as,

$$u^\mu = \frac{dx^\mu}{d\tau},$$

Covariant vectors

Perfect fluid

- *Four-velocity* is defined as,

$$u^\mu = \frac{dx^\mu}{d\tau}, \text{ where, } d\tau = \sqrt{1-\boldsymbol{v}^2}\,dt, \ \boldsymbol{v} \equiv \frac{d\boldsymbol{r}}{dt}.$$

Covariant vectors

Perfect fluid

- *Four-velocity* is defined as,

$$u^\mu = \frac{dx^\mu}{d\tau}, \text{ where, } d\tau = \sqrt{1-\boldsymbol{v}^2}\,dt, \ \boldsymbol{v} \equiv \frac{d\boldsymbol{r}}{dt}.$$

$$\Rightarrow \quad u^0 \equiv \frac{dt}{d\tau} = \gamma,$$

Covariant vectors

Perfect fluid

- *Four-velocity* is defined as,

$$u^\mu = \frac{dx^\mu}{d\tau}, \text{ where, } d\tau = \sqrt{1-\boldsymbol{v}^2}\,dt, \ \boldsymbol{v} \equiv \frac{d\boldsymbol{r}}{dt}.$$

$$\Rightarrow \quad u^0 \equiv \frac{dt}{d\tau} = \gamma, \ u^i \equiv \frac{dx^i}{d\tau}$$

Covariant vectors

Perfect fluid

- *Four-velocity* is defined as,

$$u^\mu = \frac{dx^\mu}{d\tau}, \text{ where, } d\tau = \sqrt{1-\boldsymbol{v}^2}\,dt, \ \boldsymbol{v} \equiv \frac{d\boldsymbol{r}}{dt}.$$

$$\Rightarrow \quad u^0 \equiv \frac{dt}{d\tau} = \gamma, \ u^i \equiv \frac{dx^i}{d\tau} = \frac{dx^i}{dt}\frac{dt}{d\tau} = v^i\gamma$$

Covariant vectors

Perfect fluid

- *Four-velocity* is defined as,

$$u^\mu = \frac{dx^\mu}{d\tau}, \text{ where, } d\tau = \sqrt{1-\boldsymbol{v}^2}\,dt, \ \boldsymbol{v} \equiv \frac{d\boldsymbol{r}}{dt}.$$

$$\Rightarrow \quad u^0 \equiv \frac{dt}{d\tau} = \gamma, \ u^i \equiv \frac{dx^i}{d\tau} = \frac{dx^i}{dt}\frac{dt}{d\tau} = v^i\gamma$$

- Under Lorentz transformation the *stress-energy* tensor transforms as,

$$T'^{\mu\nu} = \Lambda^\mu{}_\alpha \Lambda^\nu{}_\beta T^{\alpha\beta}.$$

Covariant vectors

Perfect fluid

- *Four-velocity* is defined as,

$$u^\mu = \frac{dx^\mu}{d\tau}, \text{ where, } d\tau = \sqrt{1-\boldsymbol{v}^2}\,dt, \ \boldsymbol{v} \equiv \frac{d\boldsymbol{r}}{dt}.$$

$$\Rightarrow \quad u^0 \equiv \frac{dt}{d\tau} = \gamma, \ u^i \equiv \frac{dx^i}{d\tau} = \frac{dx^i}{dt}\frac{dt}{d\tau} = v^i\gamma$$

- Under Lorentz transformation the *stress-energy* tensor transforms as,

$$T'^{\mu\nu} = \Lambda^\mu{}_\alpha \Lambda^\nu{}_\beta T^{\alpha\beta}.$$

- In rest frame *stress-energy* tensor of perfect fluid is given by

$$T'^{\mu\nu} = \begin{pmatrix} \epsilon & 0 & 0 & 0 \\ 0 & \mathtt{p} & 0 & 0 \\ 0 & 0 & \mathtt{p} & 0 \\ 0 & 0 & 0 & \mathtt{p} \end{pmatrix}$$

Covariant vectors

Perfect fluid

- *Four-velocity* is defined as,

$$u^\mu = \frac{dx^\mu}{d\tau}, \text{ where, } d\tau = \sqrt{1-\boldsymbol{v}^2}\,dt, \ \boldsymbol{v} \equiv \frac{d\boldsymbol{r}}{dt}.$$

$$\Rightarrow \quad u^0 \equiv \frac{dt}{d\tau} = \gamma, \ u^i \equiv \frac{dx^i}{d\tau} = \frac{dx^i}{dt}\frac{dt}{d\tau} = v^i\gamma$$

- Under Lorentz transformation the *stress-energy* tensor transforms as,

$$T'^{\mu\nu} = \Lambda^\mu{}_\alpha \Lambda^\nu{}_\beta T^{\alpha\beta}.$$

- In rest frame *stress-energy* tensor of perfect fluid is given by

$$T'^{\mu\nu} = \begin{pmatrix} \epsilon & 0 & 0 & 0 \\ 0 & \mathtt{p} & 0 & 0 \\ 0 & 0 & \mathtt{p} & 0 \\ 0 & 0 & 0 & \mathtt{p} \end{pmatrix}$$

$$T^{\mu\nu} = \Lambda_\alpha{}^\mu \Lambda_\beta{}^\nu T'^{\alpha\beta}$$

Covariant vectors

Perfect fluid

- *Four-velocity* is defined as,

$$u^\mu = \frac{dx^\mu}{d\tau}, \text{ where, } d\tau = \sqrt{1-\boldsymbol{v}^2}\,dt, \ \boldsymbol{v} \equiv \frac{d\boldsymbol{r}}{dt}.$$

$$\Rightarrow \quad u^0 \equiv \frac{dt}{d\tau} = \gamma, \ u^i \equiv \frac{dx^i}{d\tau} = \frac{dx^i}{dt}\frac{dt}{d\tau} = v^i\gamma$$

- Under Lorentz transformation the *stress-energy* tensor transforms as,

$$T'^{\mu\nu} = \Lambda^\mu{}_\alpha \Lambda^\nu{}_\beta T^{\alpha\beta}.$$

- In rest frame *stress-energy* tensor of perfect fluid is given by

$$T'^{\mu\nu} = \begin{pmatrix} \epsilon & 0 & 0 & 0 \\ 0 & \mathtt{p} & 0 & 0 \\ 0 & 0 & \mathtt{p} & 0 \\ 0 & 0 & 0 & \mathtt{p} \end{pmatrix}$$

$$\begin{aligned} T^{\mu\nu} &= \Lambda_\alpha{}^\mu \Lambda_\beta{}^\nu T'^{\alpha\beta} \\ \text{Show that, } T^{\mu\nu} &= -\mathtt{p}\eta^{\mu\nu} + (\mathtt{p}+\epsilon)u^\mu u^\nu \end{aligned}$$

Curved space time

Scalar, Vector, Tensor

- A scalar is defined by the ansatz $S'(x') = S(x)$.
- Invariant interval is given by

$$d\tau^2 = (dx^0)^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2$$

Curved space time

Scalar, Vector, Tensor

- A scalar is defined by the ansatz $S'(x') = S(x)$.
- Invariant interval is given by

$$d\tau^2 = (dx^0)^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2$$

$$\text{So, } x^\mu \rightarrow x'^\mu$$

Curved space time

Scalar, Vector, Tensor

- A scalar is defined by the ansatz $S'(x') = S(x)$.
- Invariant interval is given by

$$\begin{aligned} d\tau^2 &= (dx^0)^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2 \\ \text{So, } x^\mu &\rightarrow x'^\mu \Rightarrow dx'^\mu = \frac{\partial x'^\mu}{\partial x^\nu} dx^\nu \end{aligned}$$

Curved space time

Scalar, Vector, Tensor

- A scalar is defined by the ansatz $S'(x') = S(x)$.
- Invariant interval is given by

$$\begin{aligned} d\tau^2 &= (dx^0)^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2 \\ \text{So, } x^\mu &\rightarrow x'^\mu \Rightarrow dx'^\mu = \frac{\partial x'^\mu}{\partial x^\nu} dx^\nu \\ \Rightarrow d\tau^2 &= g_{\mu\nu} dx^\mu dx^\nu; \end{aligned}$$

Curved space time

Scalar, Vector, Tensor

- A scalar is defined by the ansatz $S'(x') = S(x)$.
- Invariant interval is given by

$$\begin{aligned} d\tau^2 &= (dx^0)^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2 \\ \text{So, } x^\mu &\rightarrow x'^\mu \Rightarrow dx'^\mu = \frac{\partial x'^\mu}{\partial x^\nu} dx^\nu \\ \Rightarrow d\tau^2 &= g_{\mu\nu} dx^\mu dx^\nu; \text{ with, } g_{\mu\nu} = g_{\nu\mu} \end{aligned}$$

Curved space time

Scalar, Vector, Tensor

- A scalar is defined by the ansatz $S'(x') = S(x)$.
- Invariant interval is given by

$$\begin{aligned} d\tau^2 &= (dx^0)^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2 \\ \text{So, } x^\mu &\rightarrow x'^\mu \Rightarrow dx'^\mu = \frac{\partial x'^\mu}{\partial x^\nu} dx^\nu \\ \Rightarrow d\tau^2 &= g_{\mu\nu} dx^\mu dx^\nu; \text{ with, } g_{\mu\nu} = g_{\nu\mu} \end{aligned}$$

- So a vector is given by the condition,

$$A'^\mu = \frac{\partial x'^\mu}{\partial x^\nu} A^\nu,$$

Curved space time

Scalar, Vector, Tensor

- A scalar is defined by the ansatz $S'(x') = S(x)$.
- Invariant interval is given by

$$\begin{aligned} d\tau^2 &= (dx^0)^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2 \\ \text{So, } x^\mu &\rightarrow x'^\mu \Rightarrow dx'^\mu = \frac{\partial x'^\mu}{\partial x^\nu} dx^\nu \\ \Rightarrow d\tau^2 &= g_{\mu\nu} dx^\mu dx^\nu; \text{ with, } g_{\mu\nu} = g_{\nu\mu} \end{aligned}$$

- So a vector is given by the condition,

$$A'^\mu = \frac{\partial x'^\mu}{\partial x^\nu} A^\nu, \; g_{\mu\nu} A^\mu A^\nu = A^2,$$

Curved space time

Scalar, Vector, Tensor

- A scalar is defined by the ansatz $S'(x') = S(x)$.
- Invariant interval is given by

$$\begin{aligned} d\tau^2 &= (dx^0)^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2 \\ \text{So, } x^\mu &\rightarrow x'^\mu \Rightarrow dx'^\mu = \frac{\partial x'^\mu}{\partial x^\nu} dx^\nu \\ \Rightarrow d\tau^2 &= g_{\mu\nu} dx^\mu dx^\nu; \text{ with, } g_{\mu\nu} = g_{\nu\mu} \end{aligned}$$

- So a vector is given by the condition,

$$A'^\mu = \frac{\partial x'^\mu}{\partial x^\nu} A^\nu, \; g_{\mu\nu} A^\mu A^\nu = A^2, \; A_\mu = g_{\mu\nu} A^\nu.$$

Curved space time

Scalar, Vector, Tensor

- A scalar is defined by the ansatz $S'(x') = S(x)$.
- Invariant interval is given by

$$\begin{aligned} d\tau^2 &= (dx^0)^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2 \\ \text{So, } x^\mu &\rightarrow x'^\mu \Rightarrow dx'^\mu = \frac{\partial x'^\mu}{\partial x^\nu} dx^\nu \\ \Rightarrow d\tau^2 &= g_{\mu\nu} dx^\mu dx^\nu; \text{ with, } g_{\mu\nu} = g_{\nu\mu} \end{aligned}$$

- So a vector is given by the condition,

$$A'^\mu = \frac{\partial x'^\mu}{\partial x^\nu} A^\nu, \; g_{\mu\nu} A^\mu A^\nu = A^2, \; A_\mu = g_{\mu\nu} A^\nu.$$

$$\Rightarrow A'_\mu B'^\mu = A_\alpha B^\alpha$$

Curved space time

Scalar, Vector, Tensor

- A scalar is defined by the ansatz $S'(x') = S(x)$.
- Invariant interval is given by

$$\begin{aligned} d\tau^2 &= (dx^0)^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2 \\ \text{So, } x^\mu &\rightarrow x'^\mu \Rightarrow dx'^\mu = \frac{\partial x'^\mu}{\partial x^\nu} dx^\nu \\ \Rightarrow d\tau^2 &= g_{\mu\nu} dx^\mu dx^\nu; \text{ with, } g_{\mu\nu} = g_{\nu\mu} \end{aligned}$$

- So a vector is given by the condition,

$$\begin{aligned} A'^\mu = \frac{\partial x'^\mu}{\partial x^\nu} A^\nu, \; g_{\mu\nu} A^\mu A^\nu &= A^2, \; A_\mu = g_{\mu\nu} A^\nu. \\ \Rightarrow A'_\mu B'^\mu = A_\alpha B^\alpha &= A_\alpha \frac{\partial x^\alpha}{\partial x'^\mu} B'^\mu \end{aligned}$$

Curved space time

Scalar, Vector, Tensor

- A scalar is defined by the ansatz $S'(x') = S(x)$.
- Invariant interval is given by

$$\begin{aligned} d\tau^2 &= (dx^0)^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2 \\ \text{So, } x^\mu &\rightarrow x'^\mu \Rightarrow dx'^\mu = \frac{\partial x'^\mu}{\partial x^\nu} dx^\nu \\ \Rightarrow d\tau^2 &= g_{\mu\nu} dx^\mu dx^\nu; \text{ with, } g_{\mu\nu} = g_{\nu\mu} \end{aligned}$$

- So a vector is given by the condition,

$$\begin{aligned} A'^\mu = \frac{\partial x'^\mu}{\partial x^\nu} A^\nu,\ g_{\mu\nu} A^\mu A^\nu &= A^2,\ A_\mu = g_{\mu\nu} A^\nu. \\ \Rightarrow A'_\mu B'^\mu = A_\alpha B^\alpha &= A_\alpha \frac{\partial x^\alpha}{\partial x'^\mu} B'^\mu \\ \Rightarrow A'_\mu &= \frac{\partial x^\nu}{\partial x'^\mu} A_\nu \end{aligned}$$

Curved space time

Scalar, Vector, Tensor

- A scalar is defined by the ansatz $S'(x') = S(x)$.
- Invariant interval is given by

$$\begin{aligned} d\tau^2 &= (dx^0)^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2 \\ \text{So, } x^\mu &\rightarrow x'^\mu \Rightarrow dx'^\mu = \frac{\partial x'^\mu}{\partial x^\nu} dx^\nu \\ \Rightarrow d\tau^2 &= g_{\mu\nu} dx^\mu dx^\nu; \text{ with, } g_{\mu\nu} = g_{\nu\mu} \end{aligned}$$

- So a vector is given by the condition,

$$\begin{aligned} A'^\mu = \frac{\partial x'^\mu}{\partial x^\nu} A^\nu, \; g_{\mu\nu} A^\mu A^\nu &= A^2, \; A_\mu = g_{\mu\nu} A^\nu. \\ \Rightarrow A'_\mu B'^\mu = A_\alpha B^\alpha &= A_\alpha \frac{\partial x^\alpha}{\partial x'^\mu} B'^\mu \\ \Rightarrow A'_\mu &= \frac{\partial x^\nu}{\partial x'^\mu} A_\nu \end{aligned}$$

- Like Minkowski metric we still have

$$g^{\alpha\mu} g_{\mu\beta} = \delta^\alpha_\beta \equiv g^\alpha_\beta.$$

Curved space time

Scalar, Vector, Tensor

- We introduce the *comma* subscript as,

$$\frac{\partial S}{\partial x'^{\mu}} = \frac{\partial x^{\nu}}{\partial x'^{\mu}} \frac{\partial S}{\partial x^{\nu}}$$

Curved space time

Scalar, Vector, Tensor

- We introduce the *comma* subscript as,

$$\frac{\partial S}{\partial x'^{\mu}} = \frac{\partial x^{\nu}}{\partial x'^{\mu}} \frac{\partial S}{\partial x^{\nu}} \Rightarrow S_{,\mu} \equiv \frac{\partial S}{\partial x^{\mu}}$$

Curved space time

Scalar, Vector, Tensor

- We introduce the *comma* subscript as,

$$\frac{\partial S}{\partial x'^{\mu}} = \frac{\partial x^{\nu}}{\partial x'^{\mu}} \frac{\partial S}{\partial x^{\nu}} \Rightarrow S_{,\mu} \equiv \frac{\partial S}{\partial x^{\mu}}$$

- A tensor of rank 2 has the following properties

$$T_{\mu}^{\ \nu} = g_{\mu\alpha} T^{\alpha\nu}, \ T^{\nu}_{\ \mu} = T^{\nu\alpha} g_{\alpha\mu}, \ T_{\mu\nu} = g_{\mu\alpha} g_{\nu\beta} T^{\alpha\beta}.$$

Curved space time

Scalar, Vector, Tensor

- We introduce the *comma* subscript as,

$$\frac{\partial S}{\partial x'^{\mu}} = \frac{\partial x^{\nu}}{\partial x'^{\mu}} \frac{\partial S}{\partial x^{\nu}} \Rightarrow S_{,\mu} \equiv \frac{\partial S}{\partial x^{\mu}}$$

- A tensor of rank 2 has the following properties

$$T_{\mu}^{\ \nu} = g_{\mu\alpha} T^{\alpha\nu}, \ T^{\nu}_{\ \mu} = T^{\nu\alpha} g_{\alpha\mu}, \ T_{\mu\nu} = g_{\mu\alpha} g_{\nu\beta} T^{\alpha\beta}.$$

- We can perform a contraction to form a scalar,

$$S = T^{\mu}_{\ \mu} = T_{\mu}^{\ \mu}$$

Curved space time

Scalar, Vector, Tensor

- We introduce the *comma* subscript as,

$$\frac{\partial S}{\partial x'^{\mu}} = \frac{\partial x^{\nu}}{\partial x'^{\mu}} \frac{\partial S}{\partial x^{\nu}} \Rightarrow S_{,\mu} \equiv \frac{\partial S}{\partial x^{\mu}}$$

- A tensor of rank 2 has the following properties

$$T_{\mu}^{\ \nu} = g_{\mu\alpha} T^{\alpha\nu}, \ T^{\nu}_{\ \mu} = T^{\nu\alpha} g_{\alpha\mu}, \ T_{\mu\nu} = g_{\mu\alpha} g_{\nu\beta} T^{\alpha\beta}.$$

- We can perform a contraction to form a scalar,

$$S = T^{\mu}_{\ \mu} = T_{\mu}^{\ \mu}$$

- Lorentz transformation of tensor in curved space time,

$$T'^{\mu}_{\ \nu} = \frac{\partial x'^{\mu}}{\partial x^{\alpha}} \frac{\partial x^{\beta}}{\partial x'^{\nu}} T^{\alpha}_{\ \beta}$$

Curved space time

Scalar, Vector, Tensor

- We introduce the *comma* subscript as,

$$\frac{\partial S}{\partial x'^{\mu}} = \frac{\partial x^{\nu}}{\partial x'^{\mu}} \frac{\partial S}{\partial x^{\nu}} \Rightarrow S_{,\mu} \equiv \frac{\partial S}{\partial x^{\mu}}$$

- A tensor of rank 2 has the following properties

$$T_{\mu}^{\ \nu} = g_{\mu\alpha} T^{\alpha\nu}, \ T^{\nu}_{\ \mu} = T^{\nu\alpha} g_{\alpha\mu}, \ T_{\mu\nu} = g_{\mu\alpha} g_{\nu\beta} T^{\alpha\beta}.$$

- We can perform a contraction to form a scalar,

$$S = T^{\mu}_{\ \mu} = T_{\mu}^{\ \mu}$$

- Lorentz transformation of tensor in curved space time,

$$T'^{\mu}_{\ \nu} = \frac{\partial x'^{\mu}}{\partial x^{\alpha}} \frac{\partial x^{\beta}}{\partial x'^{\nu}} T^{\alpha}_{\ \beta}$$

- Then metric tensor follows,

$$g'_{\alpha\beta} A'^{\alpha} A'^{\beta} = g_{\mu\nu} A^{\mu} A^{\nu}$$

Curved space time

Scalar, Vector, Tensor

- We introduce the *comma* subscript as,

$$\frac{\partial S}{\partial x'^{\mu}} = \frac{\partial x^{\nu}}{\partial x'^{\mu}} \frac{\partial S}{\partial x^{\nu}} \Rightarrow S_{,\mu} \equiv \frac{\partial S}{\partial x^{\mu}}$$

- A tensor of rank 2 has the following properties

$$T_{\mu}^{\ \nu} = g_{\mu\alpha} T^{\alpha\nu}, \ T^{\nu}_{\ \mu} = T^{\nu\alpha} g_{\alpha\mu}, \ T_{\mu\nu} = g_{\mu\alpha} g_{\nu\beta} T^{\alpha\beta}.$$

- We can perform a contraction to form a scalar,

$$S = T^{\mu}_{\ \mu} = T_{\mu}^{\ \mu}$$

- Lorentz transformation of tensor in curved space time,

$$T'^{\mu}_{\ \nu} = \frac{\partial x'^{\mu}}{\partial x^{\alpha}} \frac{\partial x^{\beta}}{\partial x'^{\nu}} T^{\alpha}_{\ \beta}$$

- Then metric tensor follows,

$$g'_{\alpha\beta} A'^{\alpha} A'^{\beta} = g_{\mu\nu} A^{\mu} A^{\nu} = g_{\mu\nu} \frac{\partial x^{\mu}}{\partial x'^{\alpha}} \frac{\partial x^{\nu}}{\partial x'^{\beta}} A'^{\alpha} A'^{\beta}$$

Curved space time

Scalar, Vector, Tensor

- We introduce the *comma* subscript as,

$$\frac{\partial S}{\partial x'^{\mu}} = \frac{\partial x^{\nu}}{\partial x'^{\mu}} \frac{\partial S}{\partial x^{\nu}} \Rightarrow S_{,\mu} \equiv \frac{\partial S}{\partial x^{\mu}}$$

- A tensor of rank 2 has the following properties

$$T_{\mu}^{\ \nu} = g_{\mu\alpha} T^{\alpha\nu}, \ T^{\nu}_{\ \mu} = T^{\nu\alpha} g_{\alpha\mu}, \ T_{\mu\nu} = g_{\mu\alpha} g_{\nu\beta} T^{\alpha\beta}.$$

- We can perform a contraction to form a scalar,

$$S = T^{\mu}_{\ \mu} = T_{\mu}^{\ \mu}$$

- Lorentz transformation of tensor in curved space time,

$$T'^{\mu}_{\ \nu} = \frac{\partial x'^{\mu}}{\partial x^{\alpha}} \frac{\partial x^{\beta}}{\partial x'^{\nu}} T^{\alpha}_{\ \beta}$$

- Then metric tensor follows,

$$g'_{\alpha\beta} A'^{\alpha} A'^{\beta} = g_{\mu\nu} A^{\mu} A^{\nu} = g_{\mu\nu} \frac{\partial x^{\mu}}{\partial x'^{\alpha}} \frac{\partial x^{\nu}}{\partial x'^{\beta}} A'^{\alpha} A'^{\beta}$$

$$g'_{\alpha\beta} = g_{\mu\nu} \frac{\partial x^{\mu}}{\partial x'^{\alpha}} \frac{\partial x^{\nu}}{\partial x'^{\beta}}$$

Curved space time

Scalar, Vector, Tensor

- We introduce the *comma* subscript as,

$$\frac{\partial S}{\partial x'^{\mu}} = \frac{\partial x^{\nu}}{\partial x'^{\mu}} \frac{\partial S}{\partial x^{\nu}} \Rightarrow S_{,\mu} \equiv \frac{\partial S}{\partial x^{\mu}}$$

- A tensor of rank 2 has the following properties

$$T_{\mu}^{\ \nu} = g_{\mu\alpha} T^{\alpha\nu}, \ T^{\nu}_{\ \mu} = T^{\nu\alpha} g_{\alpha\mu}, \ T_{\mu\nu} = g_{\mu\alpha} g_{\nu\beta} T^{\alpha\beta}.$$

- We can perform a contraction to form a scalar,

$$S = T^{\mu}_{\ \mu} = T_{\mu}^{\ \mu}$$

- Lorentz transformation of tensor in curved space time,

$$T'^{\mu}_{\ \nu} = \frac{\partial x'^{\mu}}{\partial x^{\alpha}} \frac{\partial x^{\beta}}{\partial x'^{\nu}} T^{\alpha}_{\ \beta}$$

- Then metric tensor follows,

$$g'_{\alpha\beta} A'^{\alpha} A'^{\beta} = g_{\mu\nu} A^{\mu} A^{\nu} = g_{\mu\nu} \frac{\partial x^{\mu}}{\partial x'^{\alpha}} \frac{\partial x^{\nu}}{\partial x'^{\beta}} A'^{\alpha} A'^{\beta}$$

$$g'_{\alpha\beta} = g_{\mu\nu} \frac{\partial x^{\mu}}{\partial x'^{\alpha}} \frac{\partial x^{\nu}}{\partial x'^{\beta}} \quad \& \quad g'^{\alpha\beta} = g^{\mu\nu} \frac{\partial x'^{\alpha}}{\partial x^{\mu}} \frac{\partial x'^{\beta}}{\partial x^{\nu}}$$

Principle of Equivalence

Matter-spacetime

- We almost instinctively write,

$$\boxed{m_I a = G\frac{m_G M}{R^2}}$$

even though their origins are different.

Principle of Equivalence

Matter-spacetime

- We almost instinctively write,

$$\boxed{m_I a = G\frac{m_G M}{R^2}}$$

  even though their origins are different.
- Einstein argued, $m_I = m_G$ is not a coincidence.

Principle of Equivalence

Matter-spacetime

- We almost instinctively write,

$$\boxed{m_I a = G\frac{m_G M}{R^2}}$$

  even though their origins are different.
- Einstein argued, $m_I = m_G$ is not a coincidence. This is revolutionary in the sense, space-time-matter nothing was thought to be static anymore.

Principle of Equivalence

Matter-spacetime

- We almost instinctively write,

$$\boxed{m_I a = G\frac{m_G M}{R^2}}$$

  even though their origins are different.
- Einstein argued, $m_I = m_G$ is not a coincidence. This is revolutionary in the sense, space-time-matter nothing was thought to be static anymore.
- All other forces we know, $a \propto \frac{1}{m}$.

Principle of Equivalence

Matter-spacetime

- We almost instinctively write,

$$\boxed{m_I a = G\frac{m_G M}{R^2}}$$

  even though their origins are different.
- Einstein argued, $m_I = m_G$ is not a coincidence. This is revolutionary in the sense, space-time-matter nothing was thought to be static anymore.
- All other forces we know, $a \propto \frac{1}{m}$. It is independent of mass for Gravitational force.

Principle of Equivalence

Matter-spacetime

- We almost instinctively write,

$$\boxed{m_I a = G\frac{m_G M}{R^2}}$$

  even though their origins are different.
- Einstein argued, $m_I = m_G$ is not a coincidence. This is revolutionary in the sense, space-time-matter nothing was thought to be static anymore.
- All other forces we know, $a \propto \frac{1}{m}$. It is independent of mass for Gravitational force.
- It is because "....*their motion has noting to do with their nature, it evidently has to do with the nature of the spacetime....*"

Principle of Equivalence

Matter-spacetime

- We almost instinctively write,

$$\boxed{m_I a = G\frac{m_G M}{R^2}}$$

  even though their origins are different.
- Einstein argued, $m_I = m_G$ is not a coincidence. This is revolutionary in the sense, space-time-matter nothing was thought to be static anymore.
- All other forces we know, $a \propto \frac{1}{m}$. It is independent of mass for Gravitational force.
- It is because "....*their motion has noting to do with their nature, it evidently has to do with the nature of the spacetime....*"
- **Statement:** *At every spacetime point in an arbitrary gravitational field (anytime anywhere in the universe), a local inertial (Lorentz) frame can be chosen so that the laws of the physics take on the form they have in STR.*

Curvature of space-time

Tidal gravity

- Gravitational fields are not uniform and constant. It depends on the motion of the gravitating bodies.

Curvature of space-time

Tidal gravity

- Gravitational fields are not uniform and constant. It depends on the motion of the gravitating bodies.
- Tidal gravity is the deviation from uniformity of the gravitational field at nearby points.

Curvature of space-time

Tidal gravity

- Gravitational fields are not uniform and constant. It depends on the motion of the gravitating bodies.
- Tidal gravity is the deviation from uniformity of the gravitational field at nearby points.
- Whatever the motion of a free body in an arbitrary gravitational field, it will follow a straight line trajectory over any small locality.

Curvature of space-time

Tidal gravity

- Gravitational fields are not uniform and constant. It depends on the motion of the gravitating bodies.
- Tidal gravity is the deviation from uniformity of the gravitational field at nearby points.
- Whatever the motion of a free body in an arbitrary gravitational field, it will follow a straight line trajectory over any small locality. *i.e.* Any sufficiently small region of curved spacetime is locally flat.

Curvature of space-time

Tidal gravity

- Gravitational fields are not uniform and constant. It depends on the motion of the gravitating bodies.
- Tidal gravity is the deviation from uniformity of the gravitational field at nearby points.
- Whatever the motion of a free body in an arbitrary gravitational field, it will follow a straight line trajectory over any small locality. *i.e.* Any sufficiently small region of curved spacetime is locally flat.
- **Geodesics:** The paths in curved spacetime that have the property of being locally straight are called geodesics.

Curvature of space-time

Tidal gravity

- Gravitational fields are not uniform and constant. It depends on the motion of the gravitating bodies.
- Tidal gravity is the deviation from uniformity of the gravitational field at nearby points.
- Whatever the motion of a free body in an arbitrary gravitational field, it will follow a straight line trajectory over any small locality. *i.e.* Any sufficiently small region of curved spacetime is locally flat.
- **Geodesics:** The paths in curved spacetime that have the property of being locally straight are called geodesics.
- Space time becomes curved due to gravitating bodies.

Curvature of space-time

Tidal gravity

- Gravitational fields are not uniform and constant. It depends on the motion of the gravitating bodies.
- Tidal gravity is the deviation from uniformity of the gravitational field at nearby points.
- Whatever the motion of a free body in an arbitrary gravitational field, it will follow a straight line trajectory over any small locality. *i.e.* Any sufficiently small region of curved spacetime is locally flat.
- **Geodesics:** The paths in curved spacetime that have the property of being locally straight are called geodesics.
- Space time becomes curved due to gravitating bodies. *Gravitational force* is replaced by *curvature of spacetime*.

Curvature of space-time

Tidal gravity

- Gravitational fields are not uniform and constant. It depends on the motion of the gravitating bodies.
- Tidal gravity is the deviation from uniformity of the gravitational field at nearby points.
- Whatever the motion of a free body in an arbitrary gravitational field, it will follow a straight line trajectory over any small locality. *i.e.* Any sufficiently small region of curved spacetime is locally flat.
- **Geodesics:** The paths in curved spacetime that have the property of being locally straight are called geodesics.
- Space time becomes curved due to gravitating bodies. *Gravitational force* is replaced by *curvature of spacetime*.
- In the frame of the falling body itself the spacetime is flat!!!!

Equation of motion / Geodesics

Affine connection

- In the local inertial (rest) frame (ξ) the particle moves in a straight line

$$\boxed{\frac{d^2\xi}{d\tau^2} = 0}, \text{ with, } d\tau^2 = \eta_{\alpha\beta}\, d\xi^\alpha\, d\xi^\beta$$

Equation of motion / Geodesics

Affine connection

- In the local inertial (rest) frame (ξ) the particle moves in a straight line

$$\boxed{\frac{d^2\xi}{d\tau^2} = 0}, \text{ with, } d\tau^2 = \eta_{\alpha\beta}\, d\xi^\alpha\, d\xi^\beta$$

- For an arbitrary reference frame if we have x^μ as the coordinates,

$$\frac{d}{d\tau}\left(\frac{\partial \xi^\alpha}{\partial x^\mu}\frac{dx^\mu}{d\tau}\right) = 0$$

Equation of motion / Geodesics

Affine connection

- In the local inertial (rest) frame (ξ) the particle moves in a straight line

$$\boxed{\frac{d^2\xi}{d\tau^2} = 0}, \text{ with, } d\tau^2 = \eta_{\alpha\beta}\, d\xi^\alpha\, d\xi^\beta$$

- For an arbitrary reference frame if we have x^μ as the coordinates,

$$\frac{d}{d\tau}\left(\frac{\partial \xi^\alpha}{\partial x^\mu}\frac{dx^\mu}{d\tau}\right) = 0$$

$$\Rightarrow \quad \frac{\partial \xi^\alpha}{\partial x^\mu}\frac{d^2 x^\mu}{d\tau^2} + \frac{\partial^2 \xi^\alpha}{\partial x^\mu \partial x^\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} = 0$$

Equation of motion / Geodesics

Affine connection

- In the local inertial (rest) frame (ξ) the particle moves in a straight line

$$\boxed{\frac{d^2\xi}{d\tau^2} = 0}, \text{ with, } d\tau^2 = \eta_{\alpha\beta}\, d\xi^\alpha\, d\xi^\beta$$

- For an arbitrary reference frame if we have x^μ as the coordinates,

$$\begin{aligned}
& \frac{d}{d\tau}\left(\frac{\partial \xi^\alpha}{\partial x^\mu}\frac{dx^\mu}{d\tau}\right) = 0 \\
\Rightarrow \quad & \frac{\partial \xi^\alpha}{\partial x^\mu}\frac{d^2 x^\mu}{d\tau^2} + \frac{\partial^2 \xi^\alpha}{\partial x^\mu \partial x^\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} = 0 \\
\Rightarrow \quad & \frac{\partial x^\lambda}{\partial \xi^\alpha}\left(\frac{\partial \xi^\alpha}{\partial x^\mu}\frac{d^2 x^\mu}{d\tau^2} + \frac{\partial^2 \xi^\alpha}{\partial x^\mu \partial x^\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau}\right) = 0
\end{aligned}$$

Equation of motion / Geodesics

Affine connection

- In the local inertial (rest) frame (ξ) the particle moves in a straight line

$$\boxed{\frac{d^2\xi}{d\tau^2} = 0}, \text{ with, } d\tau^2 = \eta_{\alpha\beta}\, d\xi^\alpha\, d\xi^\beta$$

- For an arbitrary reference frame if we have x^μ as the coordinates,

$$\frac{d}{d\tau}\left(\frac{\partial \xi^\alpha}{\partial x^\mu}\frac{dx^\mu}{d\tau}\right) = 0$$

$$\Rightarrow \quad \frac{\partial \xi^\alpha}{\partial x^\mu}\frac{d^2x^\mu}{d\tau^2} + \frac{\partial^2 \xi^\alpha}{\partial x^\mu \partial x^\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} = 0$$

$$\Rightarrow \quad \frac{\partial x^\lambda}{\partial \xi^\alpha}\left(\frac{\partial \xi^\alpha}{\partial x^\mu}\frac{d^2x^\mu}{d\tau^2} + \frac{\partial^2 \xi^\alpha}{\partial x^\mu \partial x^\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau}\right) = 0$$

$$\Rightarrow \quad \boxed{\frac{d^2x^\lambda}{d\tau^2} + \Gamma^\lambda_{\mu\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} = 0}$$

Equation of motion / Geodesics

Affine connection

- In the local inertial (rest) frame (ξ) the particle moves in a straight line

$$\boxed{\frac{d^2\xi}{d\tau^2} = 0}, \text{ with, } d\tau^2 = \eta_{\alpha\beta}\, d\xi^\alpha\, d\xi^\beta$$

- For an arbitrary reference frame if we have x^μ as the coordinates,

$$\frac{d}{d\tau}\left(\frac{\partial \xi^\alpha}{\partial x^\mu}\frac{dx^\mu}{d\tau}\right) = 0$$

$$\Rightarrow \quad \frac{\partial \xi^\alpha}{\partial x^\mu}\frac{d^2x^\mu}{d\tau^2} + \frac{\partial^2 \xi^\alpha}{\partial x^\mu \partial x^\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} = 0$$

$$\Rightarrow \quad \frac{\partial x^\lambda}{\partial \xi^\alpha}\left(\frac{\partial \xi^\alpha}{\partial x^\mu}\frac{d^2x^\mu}{d\tau^2} + \frac{\partial^2 \xi^\alpha}{\partial x^\mu \partial x^\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau}\right) = 0$$

$$\Rightarrow \quad \boxed{\frac{d^2x^\lambda}{d\tau^2} + \Gamma^\lambda_{\mu\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} = 0}$$

where we have used $\dfrac{dx^\lambda}{dx^\mu} = \dfrac{\partial x^\lambda}{\partial \xi^\alpha}\dfrac{\partial \xi^\alpha}{\partial x^\mu} = \delta^\lambda_\mu$,

Equation of motion / Geodesics

Affine connection

- In the local inertial (rest) frame (ξ) the particle moves in a straight line

$$\boxed{\frac{d^2\xi}{d\tau^2} = 0}, \text{ with, } d\tau^2 = \eta_{\alpha\beta}\, d\xi^\alpha\, d\xi^\beta$$

- For an arbitrary reference frame if we have x^μ as the coordinates,

$$\frac{d}{d\tau}\left(\frac{\partial \xi^\alpha}{\partial x^\mu}\frac{dx^\mu}{d\tau}\right) = 0$$

$$\Rightarrow \quad \frac{\partial \xi^\alpha}{\partial x^\mu}\frac{d^2x^\mu}{d\tau^2} + \frac{\partial^2 \xi^\alpha}{\partial x^\mu \partial x^\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} = 0$$

$$\Rightarrow \quad \frac{\partial x^\lambda}{\partial \xi^\alpha}\left(\frac{\partial \xi^\alpha}{\partial x^\mu}\frac{d^2x^\mu}{d\tau^2} + \frac{\partial^2 \xi^\alpha}{\partial x^\mu \partial x^\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau}\right) = 0$$

$$\Rightarrow \quad \boxed{\frac{d^2x^\lambda}{d\tau^2} + \Gamma^\lambda_{\mu\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} = 0}$$

where we have used $\dfrac{dx^\lambda}{dx^\mu} = \dfrac{\partial x^\lambda}{\partial \xi^\alpha}\dfrac{\partial \xi^\alpha}{\partial x^\mu} = \delta^\lambda_\mu$, and, $\Gamma^\lambda_{\mu\nu} = \dfrac{\partial x^\lambda}{\partial \xi^\alpha}\dfrac{\partial^2 \xi^\alpha}{\partial x^\mu \partial x^\nu}$

Affine connection

Transformation

- We start from the invariant proper time,

$$d\tau^2 = \eta_{\alpha\beta}\, d\xi^\alpha\, d\xi^\beta$$

Affine connection

Transformation

- We start from the invariant proper time,

$$d\tau^2 = \eta_{\alpha\beta}\, d\xi^\alpha\, d\xi^\beta = \eta_{\alpha\beta} \frac{\partial \xi^\alpha}{\partial x^\mu} dx^\mu \frac{\partial \xi^\beta}{\partial x^\nu} dx^\nu$$

Affine connection

Transformation

- We start from the invariant proper time,

$$d\tau^2 = \eta_{\alpha\beta}\, d\xi^\alpha\, d\xi^\beta = \eta_{\alpha\beta} \frac{\partial \xi^\alpha}{\partial x^\mu} dx^\mu \frac{\partial \xi^\beta}{\partial x^\nu} dx^\nu = g_{\mu\nu}\, dx^\mu\, dx^\nu ,$$

Affine connection

Transformation

- We start from the invariant proper time,

$$d\tau^2 = \eta_{\alpha\beta}\, d\xi^\alpha\, d\xi^\beta = \eta_{\alpha\beta} \frac{\partial \xi^\alpha}{\partial x^\mu} dx^\mu \frac{\partial \xi^\beta}{\partial x^\nu} dx^\nu = g_{\mu\nu}\, dx^\mu\, dx^\nu\,,\ \ g_{\mu\nu} = \frac{\partial \xi^\alpha}{\partial x^\mu} \frac{\partial \xi^\beta}{\partial x^\nu} \eta_{\alpha\beta}\,.$$

Affine connection

Transformation

- We start from the invariant proper time,

$$d\tau^2 = \eta_{\alpha\beta}\, d\xi^\alpha\, d\xi^\beta = \eta_{\alpha\beta} \frac{\partial \xi^\alpha}{\partial x^\mu} dx^\mu \frac{\partial \xi^\beta}{\partial x^\nu} dx^\nu = g_{\mu\nu}\, dx^\mu\, dx^\nu\,,\ \ g_{\mu\nu} = \frac{\partial \xi^\alpha}{\partial x^\mu} \frac{\partial \xi^\beta}{\partial x^\nu} \eta_{\alpha\beta}\,.$$

- We study few properties of affine connection as following,

$$\Gamma'^{\lambda}_{\mu\nu} = \frac{\partial x'^\lambda}{\partial \xi^\alpha} \frac{\partial^2 \xi^\alpha}{\partial x'^\mu \partial x'^\nu}$$

Affine connection

Transformation

- We start from the invariant proper time,

$$d\tau^2 = \eta_{\alpha\beta}\, d\xi^\alpha\, d\xi^\beta = \eta_{\alpha\beta} \frac{\partial \xi^\alpha}{\partial x^\mu} dx^\mu \frac{\partial \xi^\beta}{\partial x^\nu} dx^\nu = g_{\mu\nu}\, dx^\mu\, dx^\nu\,,\ \ g_{\mu\nu} = \frac{\partial \xi^\alpha}{\partial x^\mu} \frac{\partial \xi^\beta}{\partial x^\nu} \eta_{\alpha\beta}\,.$$

- We study few properties of affine connection as following,

$$\Gamma'^{\lambda}_{\mu\nu} = \frac{\partial x'^\lambda}{\partial \xi^\alpha} \frac{\partial^2 \xi^\alpha}{\partial x'^\mu \partial x'^\nu} = \frac{\partial x'^\lambda}{\partial x^\rho} \frac{\partial x^\rho}{\partial \xi^\alpha} \frac{\partial}{\partial x'^\mu} \left(\frac{\partial \xi^\alpha}{\partial x'^\nu} \right)$$

Affine connection

Transformation

- We start from the invariant proper time,

$$d\tau^2 = \eta_{\alpha\beta}\, d\xi^\alpha\, d\xi^\beta = \eta_{\alpha\beta} \frac{\partial \xi^\alpha}{\partial x^\mu} dx^\mu \frac{\partial \xi^\beta}{\partial x^\nu} dx^\nu = g_{\mu\nu}\, dx^\mu\, dx^\nu\,,\ \ g_{\mu\nu} = \frac{\partial \xi^\alpha}{\partial x^\mu} \frac{\partial \xi^\beta}{\partial x^\nu} \eta_{\alpha\beta}\,.$$

- We study few properties of affine connection as following,

$$\begin{aligned} \Gamma'^{\lambda}_{\mu\nu} &= \frac{\partial x'^\lambda}{\partial \xi^\alpha} \frac{\partial^2 \xi^\alpha}{\partial x'^\mu \partial x'^\nu} = \frac{\partial x'^\lambda}{\partial x^\rho} \frac{\partial x^\rho}{\partial \xi^\alpha} \frac{\partial}{\partial x'^\mu} \left(\frac{\partial \xi^\alpha}{\partial x'^\nu} \right) \\ &= \frac{\partial x'^\lambda}{\partial x^\rho} \frac{\partial x^\rho}{\partial \xi^\alpha} \frac{\partial}{\partial x'^\mu} \left(\frac{\partial x^\sigma}{\partial x'^\nu} \frac{\partial \xi^\alpha}{\partial x^\sigma} \right) \end{aligned}$$

Affine connection

Transformation

- We start from the invariant proper time,

$$d\tau^2 = \eta_{\alpha\beta}\, d\xi^\alpha\, d\xi^\beta = \eta_{\alpha\beta} \frac{\partial \xi^\alpha}{\partial x^\mu} dx^\mu \frac{\partial \xi^\beta}{\partial x^\nu} dx^\nu = g_{\mu\nu}\, dx^\mu\, dx^\nu, \;\; g_{\mu\nu} = \frac{\partial \xi^\alpha}{\partial x^\mu} \frac{\partial \xi^\beta}{\partial x^\nu} \eta_{\alpha\beta}.$$

- We study few properties of affine connection as following,

$$\begin{aligned}
\Gamma'^{\lambda}_{\mu\nu} &= \frac{\partial x'^\lambda}{\partial \xi^\alpha} \frac{\partial^2 \xi^\alpha}{\partial x'^\mu \partial x'^\nu} = \frac{\partial x'^\lambda}{\partial x^\rho} \frac{\partial x^\rho}{\partial \xi^\alpha} \frac{\partial}{\partial x'^\mu} \left(\frac{\partial \xi^\alpha}{\partial x'^\nu} \right) \\
&= \frac{\partial x'^\lambda}{\partial x^\rho} \frac{\partial x^\rho}{\partial \xi^\alpha} \frac{\partial}{\partial x'^\mu} \left(\frac{\partial x^\sigma}{\partial x'^\nu} \frac{\partial \xi^\alpha}{\partial x^\sigma} \right) \\
&= \frac{\partial x'^\lambda}{\partial x^\rho} \frac{\partial x^\rho}{\partial \xi^\alpha} \left[\frac{\partial x^\sigma}{\partial x'^\nu} \frac{\partial x^\tau}{\partial x'^\mu} \frac{\partial^2 \xi^\alpha}{\partial x^\tau \partial x^\sigma} + \frac{\partial^2 x^\sigma}{\partial x'^\mu \partial x'^\nu} \frac{\partial \xi^\alpha}{\partial x^\sigma} \right]
\end{aligned}$$

Affine connection

Transformation

- We start from the invariant proper time,

$$d\tau^2 = \eta_{\alpha\beta}\, d\xi^\alpha\, d\xi^\beta = \eta_{\alpha\beta} \frac{\partial \xi^\alpha}{\partial x^\mu} dx^\mu \frac{\partial \xi^\beta}{\partial x^\nu} dx^\nu = g_{\mu\nu}\, dx^\mu\, dx^\nu\,,\ \ g_{\mu\nu} = \frac{\partial \xi^\alpha}{\partial x^\mu} \frac{\partial \xi^\beta}{\partial x^\nu} \eta_{\alpha\beta}\,.$$

- We study few properties of affine connection as following,

$$\begin{aligned}
\Gamma'^{\lambda}_{\mu\nu} &= \frac{\partial x'^\lambda}{\partial \xi^\alpha} \frac{\partial^2 \xi^\alpha}{\partial x'^\mu \partial x'^\nu} = \frac{\partial x'^\lambda}{\partial x^\rho} \frac{\partial x^\rho}{\partial \xi^\alpha} \frac{\partial}{\partial x'^\mu} \left(\frac{\partial \xi^\alpha}{\partial x'^\nu} \right) \\
&= \frac{\partial x'^\lambda}{\partial x^\rho} \frac{\partial x^\rho}{\partial \xi^\alpha} \frac{\partial}{\partial x'^\mu} \left(\frac{\partial x^\sigma}{\partial x'^\nu} \frac{\partial \xi^\alpha}{\partial x^\sigma} \right) \\
&= \frac{\partial x'^\lambda}{\partial x^\rho} \frac{\partial x^\rho}{\partial \xi^\alpha} \left[\frac{\partial x^\sigma}{\partial x'^\nu} \frac{\partial x^\tau}{\partial x'^\mu} \frac{\partial^2 \xi^\alpha}{\partial x^\tau \partial x^\sigma} + \frac{\partial^2 x^\sigma}{\partial x'^\mu \partial x'^\nu} \frac{\partial \xi^\alpha}{\partial x^\sigma} \right] \\
&= \frac{\partial x'^\lambda}{\partial x^\rho} \frac{\partial x^\tau}{\partial x'^\mu} \frac{\partial x^\sigma}{\partial x'^\nu} \left(\frac{\partial x^\rho}{\partial \xi^\alpha} \frac{\partial^2 \xi^\alpha}{\partial x^\tau \partial x^\sigma} \right) + \frac{\partial x'^\lambda}{\partial x^\rho} \frac{\partial^2 x^\sigma}{\partial x'^\mu \partial x'^\nu} \delta^\rho_\sigma
\end{aligned}$$

Affine connection

Transformation

- We start from the invariant proper time,

$$d\tau^2 = \eta_{\alpha\beta}\, d\xi^\alpha\, d\xi^\beta = \eta_{\alpha\beta} \frac{\partial \xi^\alpha}{\partial x^\mu} dx^\mu \frac{\partial \xi^\beta}{\partial x^\nu} dx^\nu = g_{\mu\nu}\, dx^\mu\, dx^\nu\,, \;\; g_{\mu\nu} = \frac{\partial \xi^\alpha}{\partial x^\mu} \frac{\partial \xi^\beta}{\partial x^\nu} \eta_{\alpha\beta}\,.$$

- We study few properties of affine connection as following,

$$\begin{aligned}
\Gamma'^{\lambda}_{\mu\nu} &= \frac{\partial x'^\lambda}{\partial \xi^\alpha} \frac{\partial^2 \xi^\alpha}{\partial x'^\mu \partial x'^\nu} = \frac{\partial x'^\lambda}{\partial x^\rho} \frac{\partial x^\rho}{\partial \xi^\alpha} \frac{\partial}{\partial x'^\mu} \left(\frac{\partial \xi^\alpha}{\partial x'^\nu} \right) \\
&= \frac{\partial x'^\lambda}{\partial x^\rho} \frac{\partial x^\rho}{\partial \xi^\alpha} \frac{\partial}{\partial x'^\mu} \left(\frac{\partial x^\sigma}{\partial x'^\nu} \frac{\partial \xi^\alpha}{\partial x^\sigma} \right) \\
&= \frac{\partial x'^\lambda}{\partial x^\rho} \frac{\partial x^\rho}{\partial \xi^\alpha} \left[\frac{\partial x^\sigma}{\partial x'^\nu} \frac{\partial x^\tau}{\partial x'^\mu} \frac{\partial^2 \xi^\alpha}{\partial x^\tau \partial x^\sigma} + \frac{\partial^2 x^\sigma}{\partial x'^\mu \partial x'^\nu} \frac{\partial \xi^\alpha}{\partial x^\sigma} \right] \\
&= \frac{\partial x'^\lambda}{\partial x^\rho} \frac{\partial x^\tau}{\partial x'^\mu} \frac{\partial x^\sigma}{\partial x'^\nu} \left(\frac{\partial x^\rho}{\partial \xi^\alpha} \frac{\partial^2 \xi^\alpha}{\partial x^\tau \partial x^\sigma} \right) + \frac{\partial x'^\lambda}{\partial x^\rho} \frac{\partial^2 x^\sigma}{\partial x'^\mu \partial x'^\nu} \delta^\rho_\sigma \\
&= \frac{\partial x'^\lambda}{\partial x^\rho} \frac{\partial x^\tau}{\partial x'^\mu} \frac{\partial x^\sigma}{\partial x'^\nu} \Gamma^\rho_{\tau\sigma} + \frac{\partial x'^\lambda}{\partial x^\rho} \frac{\partial^2 x^\rho}{\partial x'^\mu \partial x'^\nu}
\end{aligned}$$

Affine connection

Christoffel symbol of second kind

- As affine connection is not a tensor, the problem is dealt in a different way. First take the derivatives of the metric,

$$\frac{\partial}{\partial x'^{\kappa}} g'_{\mu\nu} = \frac{\partial}{\partial x'^{\kappa}} \left(g_{\rho\sigma} \frac{\partial x^{\rho}}{\partial x'^{\mu}} \frac{\partial x^{\sigma}}{\partial x'^{\nu}} \right),$$

Affine connection

Christoffel symbol of second kind

- As affine connection is not a tensor, the problem is dealt in a different way. First take the derivatives of the metric,

$$\frac{\partial}{\partial x'^{\kappa}} g'_{\mu\nu} = \frac{\partial}{\partial x'^{\kappa}} \left(g_{\rho\sigma} \frac{\partial x^{\rho}}{\partial x'^{\mu}} \frac{\partial x^{\sigma}}{\partial x'^{\nu}} \right), \text{ recall, } g'^{\lambda\kappa} = g^{\alpha\beta} \frac{\partial x'^{\lambda}}{\partial x^{\alpha}} \frac{\partial x'^{\kappa}}{\partial x^{\beta}}$$

Affine connection

Christoffel symbol of second kind

- As affine connection is not a tensor, the problem is dealt in a different way. First take the derivatives of the metric,

$$\frac{\partial}{\partial x'^{\kappa}} g'_{\mu\nu} = \frac{\partial}{\partial x'^{\kappa}} \left(g_{\rho\sigma} \frac{\partial x^{\rho}}{\partial x'^{\mu}} \frac{\partial x^{\sigma}}{\partial x'^{\nu}} \right), \text{ recall, } g'^{\lambda\kappa} = g^{\alpha\beta} \frac{\partial x'^{\lambda}}{\partial x^{\alpha}} \frac{\partial x'^{\kappa}}{\partial x^{\beta}}$$

- Taking different combination one can write,

$$\begin{aligned} \frac{\partial g'_{\kappa\nu}}{\partial x'^{\mu}} + \frac{\partial g'_{\kappa\mu}}{\partial x'^{\nu}} - \frac{\partial g'_{\mu\nu}}{\partial x'^{\kappa}} &= \frac{\partial x^{\tau}}{\partial x'^{\kappa}} \frac{\partial x^{\rho}}{\partial x'^{\mu}} \frac{\partial x^{\sigma}}{\partial x'^{\nu}} \left(\frac{\partial g_{\sigma\tau}}{\partial x^{\rho}} + \frac{\partial g_{\rho\tau}}{\partial x^{\sigma}} - \frac{\partial g_{\rho\sigma}}{\partial x^{\tau}} \right) \\ &+ 2 g_{\rho\sigma} \frac{\partial x^{\sigma}}{\partial x'^{\kappa}} \frac{\partial^2 x^{\rho}}{\partial x'^{\mu} \partial x'^{\nu}} \end{aligned}$$

Affine connection

Christoffel symbol of second kind

- As affine connection is not a tensor, the problem is dealt in a different way. First take the derivatives of the metric,

$$\frac{\partial}{\partial x'^{\kappa}} g'_{\mu\nu} = \frac{\partial}{\partial x'^{\kappa}} \left(g_{\rho\sigma} \frac{\partial x^{\rho}}{\partial x'^{\mu}} \frac{\partial x^{\sigma}}{\partial x'^{\nu}} \right), \text{ recall, } g'^{\lambda\kappa} = g^{\alpha\beta} \frac{\partial x'^{\lambda}}{\partial x^{\alpha}} \frac{\partial x'^{\kappa}}{\partial x^{\beta}}$$

- Taking different combination one can write,

$$\begin{aligned} \frac{\partial g'_{\kappa\nu}}{\partial x'^{\mu}} + \frac{\partial g'_{\kappa\mu}}{\partial x'^{\nu}} - \frac{\partial g'_{\mu\nu}}{\partial x'^{\kappa}} &= \frac{\partial x^{\tau}}{\partial x'^{\kappa}} \frac{\partial x^{\rho}}{\partial x'^{\mu}} \frac{\partial x^{\sigma}}{\partial x'^{\nu}} \left(\frac{\partial g_{\sigma\tau}}{\partial x^{\rho}} + \frac{\partial g_{\rho\tau}}{\partial x^{\sigma}} - \frac{\partial g_{\rho\sigma}}{\partial x^{\tau}} \right) \\ &+ 2 g_{\rho\sigma} \frac{\partial x^{\sigma}}{\partial x'^{\kappa}} \frac{\partial^2 x^{\rho}}{\partial x'^{\mu} \partial x'^{\nu}} \end{aligned}$$

- Multiplying $\frac{1}{2}$ and then left and right sides of contravariant metric,

$$\left\{ \begin{matrix} \lambda \\ \mu\nu \end{matrix} \right\}' = \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial x^{\tau}}{\partial x'^{\mu}} \frac{\partial x^{\sigma}}{\partial x'^{\nu}} \left\{ \begin{matrix} \rho \\ \tau\sigma \end{matrix} \right\} + \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial^2 x^{\rho}}{\partial x'^{\mu} \partial x'^{\nu}},$$

Affine connection

Christoffel symbol of second kind

- As affine connection is not a tensor, the problem is dealt in a different way. First take the derivatives of the metric,

$$\frac{\partial}{\partial x'^{\kappa}} g'_{\mu\nu} = \frac{\partial}{\partial x'^{\kappa}} \left(g_{\rho\sigma} \frac{\partial x^{\rho}}{\partial x'^{\mu}} \frac{\partial x^{\sigma}}{\partial x'^{\nu}} \right), \text{ recall, } g'^{\lambda\kappa} = g^{\alpha\beta} \frac{\partial x'^{\lambda}}{\partial x^{\alpha}} \frac{\partial x'^{\kappa}}{\partial x^{\beta}}$$

- Taking different combination one can write,

$$\begin{aligned} \frac{\partial g'_{\kappa\nu}}{\partial x'^{\mu}} + \frac{\partial g'_{\kappa\mu}}{\partial x'^{\nu}} - \frac{\partial g'_{\mu\nu}}{\partial x'^{\kappa}} &= \frac{\partial x^{\tau}}{\partial x'^{\kappa}} \frac{\partial x^{\rho}}{\partial x'^{\mu}} \frac{\partial x^{\sigma}}{\partial x'^{\nu}} \left(\frac{\partial g_{\sigma\tau}}{\partial x^{\rho}} + \frac{\partial g_{\rho\tau}}{\partial x^{\sigma}} - \frac{\partial g_{\rho\sigma}}{\partial x^{\tau}} \right) \\ &+ 2 g_{\rho\sigma} \frac{\partial x^{\sigma}}{\partial x'^{\kappa}} \frac{\partial^2 x^{\rho}}{\partial x'^{\mu} \partial x'^{\nu}} \end{aligned}$$

- Multiplying $\frac{1}{2}$ and then left and right sides of contravariant metric,

$$\left\{ {\lambda \atop \mu\nu} \right\}' = \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial x^{\tau}}{\partial x'^{\mu}} \frac{\partial x^{\sigma}}{\partial x'^{\nu}} \left\{ {\rho \atop \tau\sigma} \right\} + \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial^2 x^{\rho}}{\partial x'^{\mu} \partial x'^{\nu}},$$

$$\text{with, } \left\{ {\lambda \atop \mu\nu} \right\} = \frac{1}{2} g^{\lambda\kappa} \left[\frac{\partial g_{\kappa\nu}}{\partial x^{\mu}} + \frac{\partial g_{\kappa\mu}}{\partial x^{\nu}} - \frac{\partial g_{\mu\nu}}{\partial x^{\kappa}} \right].$$

Equivalence

Christoffel symbol and Affine connection

- We have now in our hand,

$$\left\{ \begin{matrix} \lambda \\ \mu\nu \end{matrix} \right\}' = \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial x^{\tau}}{\partial x'^{\mu}} \frac{\partial x^{\sigma}}{\partial x'^{\nu}} \left\{ \begin{matrix} \rho \\ \tau\sigma \end{matrix} \right\} + \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial^2 x^{\rho}}{\partial x'^{\mu} \partial x'^{\nu}},$$

$$\Gamma'^{\lambda}_{\mu\nu} = \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial x^{\tau}}{\partial x'^{\mu}} \frac{\partial x^{\sigma}}{\partial x'^{\nu}} \Gamma^{\rho}_{\tau\sigma} + \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial^2 x^{\rho}}{\partial x'^{\mu} \partial x'^{\nu}}$$

Equivalence

Christoffel symbol and Affine connection

- We have now in our hand,

$$\left\{ \begin{matrix} \lambda \\ \mu\nu \end{matrix} \right\}' = \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial x^{\tau}}{\partial x'^{\mu}} \frac{\partial x^{\sigma}}{\partial x'^{\nu}} \left\{ \begin{matrix} \rho \\ \tau\sigma \end{matrix} \right\} + \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial^2 x^{\rho}}{\partial x'^{\mu} \partial x'^{\nu}},$$

$$\Gamma'^{\lambda}_{\mu\nu} = \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial x^{\tau}}{\partial x'^{\mu}} \frac{\partial x^{\sigma}}{\partial x'^{\nu}} \Gamma^{\rho}_{\tau\sigma} + \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial^2 x^{\rho}}{\partial x'^{\mu} \partial x'^{\nu}}$$

- So we can construct a tensor as,

$$\left[\Gamma^{\lambda}_{\mu\nu} - \left\{ \begin{matrix} \lambda \\ \mu\nu \end{matrix} \right\} \right]' = \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial x^{\tau}}{\partial x'^{\mu}} \frac{\partial x^{\sigma}}{\partial x'^{\nu}} \left[\Gamma^{\rho}_{\tau\sigma} - \left\{ \begin{matrix} \rho \\ \tau\sigma \end{matrix} \right\} \right].$$

Equivalence

Christoffel symbol and Affine connection

- We have now in our hand,

$$\left\{ \begin{matrix} \lambda \\ \mu\nu \end{matrix} \right\}' = \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial x^{\tau}}{\partial x'^{\mu}} \frac{\partial x^{\sigma}}{\partial x'^{\nu}} \left\{ \begin{matrix} \rho \\ \tau\sigma \end{matrix} \right\} + \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial^2 x^{\rho}}{\partial x'^{\mu} \partial x'^{\nu}},$$

$$\Gamma'^{\lambda}_{\mu\nu} = \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial x^{\tau}}{\partial x'^{\mu}} \frac{\partial x^{\sigma}}{\partial x'^{\nu}} \Gamma^{\rho}_{\tau\sigma} + \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial^2 x^{\rho}}{\partial x'^{\mu} \partial x'^{\nu}}$$

- So we can construct a tensor as,

$$\left[\Gamma^{\lambda}_{\mu\nu} - \left\{ \begin{matrix} \lambda \\ \mu\nu \end{matrix} \right\} \right]' = \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial x^{\tau}}{\partial x'^{\mu}} \frac{\partial x^{\sigma}}{\partial x'^{\nu}} \left[\Gamma^{\rho}_{\tau\sigma} - \left\{ \begin{matrix} \rho \\ \tau\sigma \end{matrix} \right\} \right].$$

- Equivalence principle $\Rightarrow$ there is always a flat inertial frame ξ^{α} present.

Equivalence

Christoffel symbol and Affine connection

- We have now in our hand,

$$\begin{Bmatrix} \lambda \\ \mu\nu \end{Bmatrix}' = \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial x^{\tau}}{\partial x'^{\mu}} \frac{\partial x^{\sigma}}{\partial x'^{\nu}} \begin{Bmatrix} \rho \\ \tau\sigma \end{Bmatrix} + \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial^2 x^{\rho}}{\partial x'^{\mu} \partial x'^{\nu}},$$

$$\Gamma'^{\lambda}_{\mu\nu} = \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial x^{\tau}}{\partial x'^{\mu}} \frac{\partial x^{\sigma}}{\partial x'^{\nu}} \Gamma^{\rho}_{\tau\sigma} + \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial^2 x^{\rho}}{\partial x'^{\mu} \partial x'^{\nu}}$$

- So we can construct a tensor as,

$$\left[\Gamma^{\lambda}_{\mu\nu} - \begin{Bmatrix} \lambda \\ \mu\nu \end{Bmatrix} \right]' = \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial x^{\tau}}{\partial x'^{\mu}} \frac{\partial x^{\sigma}}{\partial x'^{\nu}} \left[\Gamma^{\rho}_{\tau\sigma} - \begin{Bmatrix} \rho \\ \tau\sigma \end{Bmatrix} \right].$$

- Equivalence principle $\Rightarrow$ there is always a flat inertial frame ξ^{α} present. $\Rightarrow \Gamma^{\lambda}_{\mu\nu} = 0$,

Equivalence

Christoffel symbol and Affine connection

- We have now in our hand,

$$\begin{Bmatrix} \lambda \\ \mu\nu \end{Bmatrix}' = \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial x^{\tau}}{\partial x'^{\mu}} \frac{\partial x^{\sigma}}{\partial x'^{\nu}} \begin{Bmatrix} \rho \\ \tau\sigma \end{Bmatrix} + \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial^2 x^{\rho}}{\partial x'^{\mu} \partial x'^{\nu}},$$

$$\Gamma'^{\lambda}_{\mu\nu} = \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial x^{\tau}}{\partial x'^{\mu}} \frac{\partial x^{\sigma}}{\partial x'^{\nu}} \Gamma^{\rho}_{\tau\sigma} + \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial^2 x^{\rho}}{\partial x'^{\mu} \partial x'^{\nu}}$$

- So we can construct a tensor as,

$$\left[\Gamma^{\lambda}_{\mu\nu} - \begin{Bmatrix} \lambda \\ \mu\nu \end{Bmatrix} \right]' = \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial x^{\tau}}{\partial x'^{\mu}} \frac{\partial x^{\sigma}}{\partial x'^{\nu}} \left[\Gamma^{\rho}_{\tau\sigma} - \begin{Bmatrix} \rho \\ \tau\sigma \end{Bmatrix} \right].$$

- Equivalence principle $\Rightarrow$ there is always a flat inertial frame ξ^{α} present. $\Rightarrow \Gamma^{\lambda}_{\mu\nu} = 0$, as derivative of metric vanishes, Christoffel symbols also vanish.

Equivalence

Christoffel symbol and Affine connection

- We have now in our hand,

$$\begin{Bmatrix} \lambda \\ \mu\nu \end{Bmatrix}' = \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial x^{\tau}}{\partial x'^{\mu}} \frac{\partial x^{\sigma}}{\partial x'^{\nu}} \begin{Bmatrix} \rho \\ \tau\sigma \end{Bmatrix} + \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial^2 x^{\rho}}{\partial x'^{\mu} \partial x'^{\nu}},$$

$$\Gamma'^{\lambda}_{\mu\nu} = \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial x^{\tau}}{\partial x'^{\mu}} \frac{\partial x^{\sigma}}{\partial x'^{\nu}} \Gamma^{\rho}_{\tau\sigma} + \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial^2 x^{\rho}}{\partial x'^{\mu} \partial x'^{\nu}}$$

- So we can construct a tensor as,

$$\left[\Gamma^{\lambda}_{\mu\nu} - \begin{Bmatrix} \lambda \\ \mu\nu \end{Bmatrix} \right]' = \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial x^{\tau}}{\partial x'^{\mu}} \frac{\partial x^{\sigma}}{\partial x'^{\nu}} \left[\Gamma^{\rho}_{\tau\sigma} - \begin{Bmatrix} \rho \\ \tau\sigma \end{Bmatrix} \right].$$

- Equivalence principle $\Rightarrow$ there is always a flat inertial frame ξ^{α} present. $\Rightarrow \Gamma^{\lambda}_{\mu\nu} = 0$, as derivative of metric vanishes, Christoffel symbols also vanish. Now, difference between $\Gamma^{\lambda}_{\mu\nu}$ and Christoffel symbol vanishes in this ref. frame, which is a tensor.

Equivalence

Christoffel symbol and Affine connection

- We have now in our hand,

$$\left\{ \begin{matrix} \lambda \\ \mu\nu \end{matrix} \right\}' = \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial x^{\tau}}{\partial x'^{\mu}} \frac{\partial x^{\sigma}}{\partial x'^{\nu}} \left\{ \begin{matrix} \rho \\ \tau\sigma \end{matrix} \right\} + \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial^2 x^{\rho}}{\partial x'^{\mu} \partial x'^{\nu}},$$

$$\Gamma'^{\lambda}_{\mu\nu} = \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial x^{\tau}}{\partial x'^{\mu}} \frac{\partial x^{\sigma}}{\partial x'^{\nu}} \Gamma^{\rho}_{\tau\sigma} + \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial^2 x^{\rho}}{\partial x'^{\mu} \partial x'^{\nu}}$$

- So we can construct a tensor as,

$$\left[\Gamma^{\lambda}_{\mu\nu} - \left\{ \begin{matrix} \lambda \\ \mu\nu \end{matrix} \right\} \right]' = \frac{\partial x'^{\lambda}}{\partial x^{\rho}} \frac{\partial x^{\tau}}{\partial x'^{\mu}} \frac{\partial x^{\sigma}}{\partial x'^{\nu}} \left[\Gamma^{\rho}_{\tau\sigma} - \left\{ \begin{matrix} \rho \\ \tau\sigma \end{matrix} \right\} \right].$$

- Equivalence principle $\Rightarrow$ there is always a flat inertial frame ξ^{α} present. $\Rightarrow \Gamma^{\lambda}_{\mu\nu} = 0$, as derivative of metric vanishes, Christoffel symbols also vanish. Now, difference between $\Gamma^{\lambda}_{\mu\nu}$ and Christoffel symbol vanishes in this ref. frame, which is a tensor. So it vanishes in all inertial frames.

Equivalence

Christoffel symbol and Affine connection

- So we have,

$$\Gamma^{\lambda}_{\mu\nu} = \left\{ \begin{matrix} \lambda \\ \mu\nu \end{matrix} \right\} = \frac{1}{2} g^{\lambda\kappa} \left(g_{\kappa\nu,\mu} + g_{\kappa\mu,\nu} - g_{\mu\nu,\kappa} \right)$$

Equivalence

Christoffel symbol and Affine connection

- So we have,

$$\Gamma^{\lambda}_{\mu\nu} = \left\{ \begin{matrix} \lambda \\ \mu\nu \end{matrix} \right\} = \frac{1}{2} g^{\lambda\kappa} \left(g_{\kappa\nu,\mu} + g_{\kappa\mu,\nu} - g_{\mu\nu,\kappa} \right)$$

- It is useful sometimes to lower the index of $\Gamma^{\lambda}_{\mu\nu}$ as

$$\Gamma_{\kappa\mu\nu} = g_{\kappa\lambda} \Gamma^{\lambda}_{\mu\nu}$$

Equivalence

Christoffel symbol and Affine connection

- So we have,

$$\Gamma^{\lambda}_{\mu\nu} = \left\{ \begin{matrix} \lambda \\ \mu\nu \end{matrix} \right\} = \frac{1}{2} g^{\lambda\kappa} \left(g_{\kappa\nu,\mu} + g_{\kappa\mu,\nu} - g_{\mu\nu,\kappa} \right)$$

- It is useful sometimes to lower the index of $\Gamma^{\lambda}_{\mu\nu}$ as

$$\Gamma_{\kappa\mu\nu} = g_{\kappa\lambda} \Gamma^{\lambda}_{\mu\nu}$$

- With this the Christoffel symbol of first kind is defined as,

$$\Gamma_{\kappa\mu\nu} = \left[\begin{matrix} \kappa \\ \mu\nu \end{matrix} \right] = \frac{1}{2} \left(g_{\kappa\nu,\mu} + g_{\kappa\mu,\nu} - g_{\mu\nu,\kappa} \right)$$

Equivalence

Christoffel symbol and Affine connection

- So we have,

$$\Gamma^{\lambda}_{\mu\nu} = \left\{ \begin{matrix} \lambda \\ \mu\nu \end{matrix} \right\} = \frac{1}{2} g^{\lambda\kappa} \left(g_{\kappa\nu,\mu} + g_{\kappa\mu,\nu} - g_{\mu\nu,\kappa} \right)$$

- It is useful sometimes to lower the index of $\Gamma^{\lambda}_{\mu\nu}$ as

$$\Gamma_{\kappa\mu\nu} = g_{\kappa\lambda} \Gamma^{\lambda}_{\mu\nu}$$

- With this the Christoffel symbol of first kind is defined as,

$$\Gamma_{\kappa\mu\nu} = \left[\begin{matrix} \kappa \\ \mu\nu \end{matrix} \right] = \frac{1}{2} \left(g_{\kappa\nu,\mu} + g_{\kappa\mu,\nu} - g_{\mu\nu,\kappa} \right)$$

- Show that

$$\Gamma_{\kappa\mu\nu} + \Gamma_{\mu\kappa\nu} = \mathrm{g}_{\mu\kappa,\nu}.$$

Comparison with Newtonian gravity

Static field

- Metric takes the place of gravitational constant of Newtonian gravity.

Comparison with Newtonian gravity

Static field

- Metric takes the place of gravitational constant of Newtonian gravity.
- For a slowly moving particle in a weak static field

$$d\tau = \left(dt^2 - d\boldsymbol{r}^2\right)^{1/2}$$

Comparison with Newtonian gravity

Static field

- Metric takes the place of gravitational constant of Newtonian gravity.
- For a slowly moving particle in a weak static field

$$d\tau = \left(dt^2 - d\boldsymbol{r}^2\right)^{1/2} = \left(1 - \boldsymbol{v}^2\right)^{1/2} dt, \text{ where, } \boldsymbol{v} = \frac{d\boldsymbol{r}}{dt}.$$

Comparison with Newtonian gravity

Static field

- Metric takes the place of gravitational constant of Newtonian gravity.
- For a slowly moving particle in a weak static field

$$d\tau = \left(dt^2 - d\boldsymbol{r}^2\right)^{1/2} = \left(1 - \boldsymbol{v}^2\right)^{1/2} dt, \text{ where, } \boldsymbol{v} = \frac{d\boldsymbol{r}}{dt}.$$

The slow moving assumption signifies,

$$\frac{d\boldsymbol{r}}{d\tau} << \frac{dt}{d\tau} \approx 1.$$

Comparison with Newtonian gravity

Static field

- Metric takes the place of gravitational constant of Newtonian gravity.
- For a slowly moving particle in a weak static field

$$d\tau = \left(dt^2 - d\boldsymbol{r}^2\right)^{1/2} = \left(1 - \boldsymbol{v}^2\right)^{1/2} dt, \text{ where, } \boldsymbol{v} = \frac{d\boldsymbol{r}}{dt}.$$

The slow moving assumption signifies,

$$\frac{d\boldsymbol{r}}{d\tau} << \frac{dt}{d\tau} \approx 1.$$

- The Geodesic equation is given as,

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{\nu\rho} \frac{dx^\nu}{d\tau} \frac{dx^\rho}{d\tau} = 0$$

Comparison with Newtonian gravity

Static field

- Metric takes the place of gravitational constant of Newtonian gravity.
- For a slowly moving particle in a weak static field

$$d\tau = \left(dt^2 - d\boldsymbol{r}^2\right)^{1/2} = \left(1 - \boldsymbol{v}^2\right)^{1/2} dt, \text{ where, } \boldsymbol{v} = \frac{d\boldsymbol{r}}{dt}.$$

The slow moving assumption signifies,

$$\frac{d\boldsymbol{r}}{d\tau} << \frac{dt}{d\tau} \approx 1.$$

- The Geodesic equation is given as,

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{\nu\rho} \frac{dx^\nu}{d\tau} \frac{dx^\rho}{d\tau} = 0 \Rightarrow \frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{00} \left(\frac{dt}{d\tau}\right)^2 = 0$$

Comparison with Newtonian gravity

Static field

- Metric takes the place of gravitational constant of Newtonian gravity.
- For a slowly moving particle in a weak static field

$$d\tau = \left(dt^2 - d\boldsymbol{r}^2\right)^{1/2} = \left(1 - \boldsymbol{v}^2\right)^{1/2} dt, \text{ where, } \boldsymbol{v} = \frac{d\boldsymbol{r}}{dt}.$$

The slow moving assumption signifies,

$$\frac{d\boldsymbol{r}}{d\tau} << \frac{dt}{d\tau} \approx 1.$$

- The Geodesic equation is given as,

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{\nu\rho} \frac{dx^\nu}{d\tau} \frac{dx^\rho}{d\tau} = 0 \Rightarrow \frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{00} \left(\frac{dt}{d\tau}\right)^2 = 0$$

- As the field is static, all the time derivatives of metric vanishes $\Rightarrow$

$$\Gamma^\mu_{00} = \frac{1}{2} g^{\mu\nu} \left(2 g_{\nu 0,0} - g_{00,\nu}\right)$$

Comparison with Newtonian gravity

Static field

- Metric takes the place of gravitational constant of Newtonian gravity.
- For a slowly moving particle in a weak static field

$$d\tau = \left(dt^2 - d\boldsymbol{r}^2\right)^{1/2} = \left(1 - \boldsymbol{v}^2\right)^{1/2} dt, \text{ where, } \boldsymbol{v} = \frac{d\boldsymbol{r}}{dt}.$$

The slow moving assumption signifies,

$$\frac{d\boldsymbol{r}}{d\tau} << \frac{dt}{d\tau} \approx 1.$$

- The Geodesic equation is given as,

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{\nu\rho} \frac{dx^\nu}{d\tau} \frac{dx^\rho}{d\tau} = 0 \Rightarrow \frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{00} \left(\frac{dt}{d\tau}\right)^2 = 0$$

- As the field is static, all the time derivatives of metric vanishes $\Rightarrow$

$$\Gamma^\mu_{00} = \frac{1}{2} g^{\mu\nu} \left(2 g_{\nu 0,0} - g_{00,\nu}\right) = -\frac{1}{2} g^{\mu\nu} g_{00,\nu},$$

Comparison with Newtonian gravity

Static field

- Metric takes the place of gravitational constant of Newtonian gravity.
- For a slowly moving particle in a weak static field

$$d\tau = \left(dt^2 - d\boldsymbol{r}^2\right)^{1/2} = \left(1 - \boldsymbol{v}^2\right)^{1/2} dt, \text{ where, } \boldsymbol{v} = \frac{d\boldsymbol{r}}{dt}.$$

The slow moving assumption signifies,

$$\frac{d\boldsymbol{r}}{d\tau} << \frac{dt}{d\tau} \approx 1.$$

- The Geodesic equation is given as,

$$\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{\nu\rho}\frac{dx^\nu}{d\tau}\frac{dx^\rho}{d\tau} = 0 \Rightarrow \frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{00}\left(\frac{dt}{d\tau}\right)^2 = 0$$

- As the field is static, all the time derivatives of metric vanishes $\Rightarrow$

$$\Gamma^\mu_{00} = \frac{1}{2}g^{\mu\nu}\left(2g_{\nu 0,0} - g_{00,\nu}\right) = -\frac{1}{2}g^{\mu\nu}g_{00,\nu}, \text{ Now, } g_{00} = (1+\delta)\eta_{00}$$

Comparison with Newtonian gravity

Static field

- Metric takes the place of gravitational constant of Newtonian gravity.
- For a slowly moving particle in a weak static field

$$d\tau = \left(dt^2 - d\boldsymbol{r}^2\right)^{1/2} = \left(1 - \boldsymbol{v}^2\right)^{1/2} dt, \text{ where, } \boldsymbol{v} = \frac{d\boldsymbol{r}}{dt}.$$

The slow moving assumption signifies,

$$\frac{d\boldsymbol{r}}{d\tau} << \frac{dt}{d\tau} \approx 1.$$

- The Geodesic equation is given as,

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{\nu\rho} \frac{dx^\nu}{d\tau} \frac{dx^\rho}{d\tau} = 0 \Rightarrow \frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{00} \left(\frac{dt}{d\tau}\right)^2 = 0$$

- As the field is static, all the time derivatives of metric vanishes $\Rightarrow$

$$\Gamma^\mu_{00} = \frac{1}{2} g^{\mu\nu} \left(2 g_{\nu 0,0} - g_{00,\nu}\right) = -\frac{1}{2} g^{\mu\nu} g_{00,\nu}, \text{ Now, } g_{00} = (1+\delta)\eta_{00}$$

$$\Rightarrow \quad \Gamma^\mu_{00} = -\frac{1}{2} \eta^{\mu\nu} \eta_{00} \frac{d\delta}{dx^\nu}$$

Comparison with Newtonian gravity

Static field

- Metric takes the place of gravitational constant of Newtonian gravity.
- For a slowly moving particle in a weak static field

$$d\tau = \left(dt^2 - d\boldsymbol{r}^2\right)^{1/2} = \left(1 - \boldsymbol{v}^2\right)^{1/2} dt, \text{ where, } \boldsymbol{v} = \frac{d\boldsymbol{r}}{dt}.$$

The slow moving assumption signifies,

$$\frac{d\boldsymbol{r}}{d\tau} << \frac{dt}{d\tau} \approx 1.$$

- The Geodesic equation is given as,

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{\nu\rho}\frac{dx^\nu}{d\tau}\frac{dx^\rho}{d\tau} = 0 \Rightarrow \frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{00}\left(\frac{dt}{d\tau}\right)^2 = 0$$

- As the field is static, all the time derivatives of metric vanishes $\Rightarrow$

$$\Gamma^\mu_{00} = \frac{1}{2}g^{\mu\nu}\left(2g_{\nu 0,0} - g_{00,\nu}\right) = -\frac{1}{2}g^{\mu\nu}g_{00,\nu}, \text{ Now, } g_{00} = (1+\delta)\eta_{00}$$

$$\Rightarrow \quad \Gamma^\mu_{00} = -\frac{1}{2}\eta^{\mu\nu}\eta_{00}\frac{d\delta}{dx^\nu} \Rightarrow \Gamma^0_{00} = 0, \ \Gamma^i_{00} = \frac{1}{2}\frac{d\delta}{dx^i}$$

Comparison with Newtonian gravity

Static field

- The geodesic equations are finally written as

$$\frac{d^2\boldsymbol{r}}{d\tau^2} = -\frac{1}{2}\left(\frac{dt}{d\tau}\right)^2 \nabla\delta, \qquad \frac{d^2 t}{d\tau^2} = 0$$

Comparison with Newtonian gravity

Static field

- The geodesic equations are finally written as

$$\frac{d^2\boldsymbol{r}}{d\tau^2} = -\frac{1}{2}\left(\frac{dt}{d\tau}\right)^2 \nabla\delta, \qquad \frac{d^2 t}{d\tau^2} = 0$$

$$\Rightarrow \quad \frac{d^2\boldsymbol{r}}{dt^2}\left(\frac{dt}{d\tau}\right)^2 + \frac{d\boldsymbol{r}}{dt}\frac{d^2 t}{dt^2} = -\frac{1}{2}\left(\frac{dt}{d\tau}\right)^2 \nabla\delta$$

Comparison with Newtonian gravity

Static field

- The geodesic equations are finally written as

$$\frac{d^2\boldsymbol{r}}{d\tau^2} = -\frac{1}{2}\left(\frac{dt}{d\tau}\right)^2 \nabla\delta, \qquad \frac{d^2 t}{d\tau^2} = 0$$

$$\Rightarrow \quad \frac{d^2\boldsymbol{r}}{dt^2}\left(\frac{dt}{d\tau}\right)^2 + \frac{d\boldsymbol{r}}{dt}\frac{d^2 t}{dt^2} = -\frac{1}{2}\left(\frac{dt}{d\tau}\right)^2 \nabla\delta$$

$$\Rightarrow \quad \frac{d^2\boldsymbol{r}}{dt^2} = -\frac{1}{2}\nabla\delta,$$

Comparison with Newtonian gravity

Static field

- The geodesic equations are finally written as

$$\frac{d^2\boldsymbol{r}}{d\tau^2} = -\frac{1}{2}\left(\frac{dt}{d\tau}\right)^2 \nabla\delta, \qquad \frac{d^2 t}{d\tau^2} = 0$$

$$\Rightarrow \quad \frac{d^2\boldsymbol{r}}{dt^2}\left(\frac{dt}{d\tau}\right)^2 + \frac{d\boldsymbol{r}}{dt}\frac{d^2 t}{dt^2} = -\frac{1}{2}\left(\frac{dt}{d\tau}\right)^2 \nabla\delta$$

$$\Rightarrow \quad \frac{d^2\boldsymbol{r}}{dt^2} = -\frac{1}{2}\nabla\delta, \qquad \text{Newtonian gravity,} \quad \frac{d^2\boldsymbol{r}}{dt^2} = -\nabla V$$

Comparison with Newtonian gravity

Static field

- The geodesic equations are finally written as

$$\frac{d^2\boldsymbol{r}}{d\tau^2} = -\frac{1}{2}\left(\frac{dt}{d\tau}\right)^2 \nabla\delta, \qquad \frac{d^2 t}{d\tau^2} = 0$$

$$\Rightarrow \quad \frac{d^2\boldsymbol{r}}{dt^2}\left(\frac{dt}{d\tau}\right)^2 + \frac{d\boldsymbol{r}}{dt}\frac{d^2 t}{dt^2} = -\frac{1}{2}\left(\frac{dt}{d\tau}\right)^2 \nabla\delta$$

$$\Rightarrow \quad \frac{d^2\boldsymbol{r}}{dt^2} = -\frac{1}{2}\nabla\delta, \qquad \text{Newtonian gravity, } \frac{d^2\boldsymbol{r}}{dt^2} = -\nabla V$$

$$\Rightarrow \quad \delta = 2V$$

Comparison with Newtonian gravity

Static field

- The geodesic equations are finally written as

$$\frac{d^2\boldsymbol{r}}{d\tau^2} = -\frac{1}{2}\left(\frac{dt}{d\tau}\right)^2 \nabla\delta, \qquad \frac{d^2 t}{d\tau^2} = 0$$

$$\Rightarrow \quad \frac{d^2\boldsymbol{r}}{dt^2}\left(\frac{dt}{d\tau}\right)^2 + \frac{d\boldsymbol{r}}{dt}\frac{d^2 t}{dt^2} = -\frac{1}{2}\left(\frac{dt}{d\tau}\right)^2 \nabla\delta$$

$$\Rightarrow \quad \frac{d^2\boldsymbol{r}}{dt^2} = -\frac{1}{2}\nabla\delta, \qquad \text{Newtonian gravity,} \quad \frac{d^2\boldsymbol{r}}{dt^2} = -\nabla V$$

$$\Rightarrow \quad \delta = 2V \Rightarrow g_{00} = 1 + 2V$$

Comparison with Newtonian gravity

Static field

- The geodesic equations are finally written as

$$\frac{d^2\boldsymbol{r}}{d\tau^2} = -\frac{1}{2}\left(\frac{dt}{d\tau}\right)^2 \nabla\delta, \qquad \frac{d^2 t}{d\tau^2} = 0$$

$$\Rightarrow \quad \frac{d^2\boldsymbol{r}}{dt^2}\left(\frac{dt}{d\tau}\right)^2 + \frac{d\boldsymbol{r}}{dt}\frac{d^2 t}{dt^2} = -\frac{1}{2}\left(\frac{dt}{d\tau}\right)^2 \nabla\delta$$

$$\Rightarrow \quad \frac{d^2\boldsymbol{r}}{dt^2} = -\frac{1}{2}\nabla\delta, \qquad \text{Newtonian gravity,} \quad \frac{d^2\boldsymbol{r}}{dt^2} = -\nabla V$$

$$\Rightarrow \quad \delta = 2V \Rightarrow g_{00} = 1 + 2V = 1 - 2\frac{GM}{r}$$

Comparison with Newtonian gravity

Static field

- The geodesic equations are finally written as

$$\frac{d^2\boldsymbol{r}}{d\tau^2} = -\frac{1}{2}\left(\frac{dt}{d\tau}\right)^2 \nabla\delta, \qquad \frac{d^2 t}{d\tau^2} = 0$$

$$\Rightarrow \quad \frac{d^2\boldsymbol{r}}{dt^2}\left(\frac{dt}{d\tau}\right)^2 + \frac{d\boldsymbol{r}}{dt}\frac{d^2 t}{dt^2} = -\frac{1}{2}\left(\frac{dt}{d\tau}\right)^2 \nabla\delta$$

$$\Rightarrow \quad \frac{d^2\boldsymbol{r}}{dt^2} = -\frac{1}{2}\nabla\delta, \qquad \text{Newtonian gravity, } \frac{d^2\boldsymbol{r}}{dt^2} = -\nabla V$$

$$\Rightarrow \quad \delta = 2V \Rightarrow g_{00} = 1 + 2V = 1 - 2\frac{GM}{r}$$

This is the only independent term in Newtonian limit.

References:

- Compact Stars, Springer, **Norman K. Glendenning**.
- Quantum Field Theory, Cambridge University Press, **Lewis H. Ryder**.